Who Owns the Number?
A Pre-Registered Audit of Five Owners Behind AI-Agent Attack Success

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***Abstract*—A leaderboard attack-success number reads as a property of the model. It is a property of the contract. YHZ measures its five owners: admission rule, information contract, session wrapper, compute allowance, execution evidence. We move one at a time under pre-registration, every other byte frozen, so each number belongs to an owner. Cryptography is the setting, not the claim.**

On matched tasks one sentence decides. State the zero-reference premise and nearly every Claude- and GPT-family session recovers the plaintext on turn one; withhold it and success collapses, one-way in all sixteen paired cohorts (fifteen Holm-significant, FWER 0.05). Our own undeclared rule moves the same fixed programs 32 points; an output-preserving cleanup lands at $+3.9$ against the declared -0.8, so the movement was the rule's. The wrapper is a null: advantage bounded at 10.7% (Wilson 95%).

Outside our harness, three published graders lift correct programs from zero to 18 of 282. A published protocol at its declared 12,000-token allowance leaves 30 of 60 sessions with no action, 10 provider-filtered, so 20 of 50 solve; raising it gives 28 of 50 and 8 of 10 pairs, a 16-point swing the budget owns. Seventeen slots never returned output: unmeasured, not failures.

None of this grades an agent; it prices the contract. An out-of-domain audit-forensics task reproduces the premise effect (8/8 vs 0/8), five preconditions (the YHZ floor) complete the standard, and the share shrinks as models strengthen: the method travels, the magnitude does not. Frozen cohorts and audit trail are public.

***Index Terms*—AI agents, cryptographic attacks, exploit construction, evaluation validity, benchmark integrity, evaluation methodology, agentic AI security, pre-registration, information contract, evaluator sensitivity, verification, provenance**

1. Introduction

An agent that solves 40% of a security benchmark is assumed to be the most capable one. This paper shows that the 40% may have less to do with the agent than with who wrote the prompt, who wrote the evaluator, and what each chose to disclose. A reported success is jointly produced by identifiable owners: the agent's program, the evaluator's admission rule, the prompt's information contract, and the session wrapper (Figure 1); behind them stands a further question, whether the retained evidence credits the claimed execution. Move an owner and the number can move with the agent unchanged — and where it does not, the null is measured and registered. YHZ is a measurement instrument built on that observation: matched cryptographic recovery tasks on which each owner can be intervened on separately, every design pre-specified and hash-frozen. The number audited end-to-end is the prior YHZ benchmark's own reported contrast [1] (Section 2); no third-party published number is regraded end-to-end — their releases ship no records to regrade — and the published NYU CTF protocol is audited by registered reproduction instead (Section 7). The stakes are operational: these numbers gate security procurement, justify deploying an autonomous agent into a security-operations pipeline, and assign accountability when the agent acts — each gate inherits whichever owner the benchmark never measured.

Why a cryptographic testbed. The contribution is the ownership-audit method — matched, hash-frozen interventions that move one owner at a time — not cryptography. Cryptography is the testbed because its verdicts are ground truth: admission, naming, and premise slots each admit an intervention while target, oracle, seeds, and budget stay byte-identical; a contested success definition would import the very discretion this audit measures. The design transfers to domains with a binarizable oracle verdict and a load-bearing task premise (the transfer rule: Section 10.2) — a first out-of-domain witness, not a universal claim.

Three measured shares anchor the claim. *Admission:* with program bytes unchanged, the evaluator in force holds the representation contrast (base64 vs. hexadecimal observations) at -0.8. A narrower filter moves it by over thirty points — a within-harness counterfactual (the external regrade moved the contrast on 0 of 2 defined tasks; undefined on the other 2). An output-preserving cleanup returns it past zero (Section 4).

Information contract: a pre-specified and hash-frozen 2×2 freezes observations, target, seeds, budget, and interface, and isolates one load-bearing slot — the reference premise — across four labels and a fresh-seed $n{=}16$ cohort. Further fixtures bracket the claim: a second hidden-premise family (AES-CTR) where both withheld arms read 0/16, and an observable-premise control (ECDSA) returning the pre-registered null (32/32 withheld). Three modern-AEAD fixtures (AES-GCM, ChaCha20-Poly1305, AES-GCM-SIV) repeat the collapse on OS-CSPRNG-sampled instances, the last by design accepting nothing. The identical cell succeeds 4/4 on RC4, a hash-frozen honest null (Section 5).

The wave's last two cohorts separate two readings of the same withheld arm. On fresh CSPRNG seeds at 96 instances the paired contrast reads 95/0 ($p = 5.0 \times 10^{-29}$). With the budget doubled and the transport repaired, one label answers every session and every withheld answer is wrong; a second label's withheld arm is budget-bound rather than unmeasured (Section 5).

Wrapper: across ten matched pairs, acceptance is descriptively indistinguishable between interactive and one-shot wrappers (5/10 vs 7/10; two-sided 0.65), and the hash-frozen all-measures falsifier is FALSE via the revisions clause (2 recoveries on 7 rejections). The pre-registered powered 32-pair cohort then resolves the question at its registered bar: 0/32 discordances in each direction, the interactive arm's advantage bounded at 10.7% (Wilson 95%; Section 6).

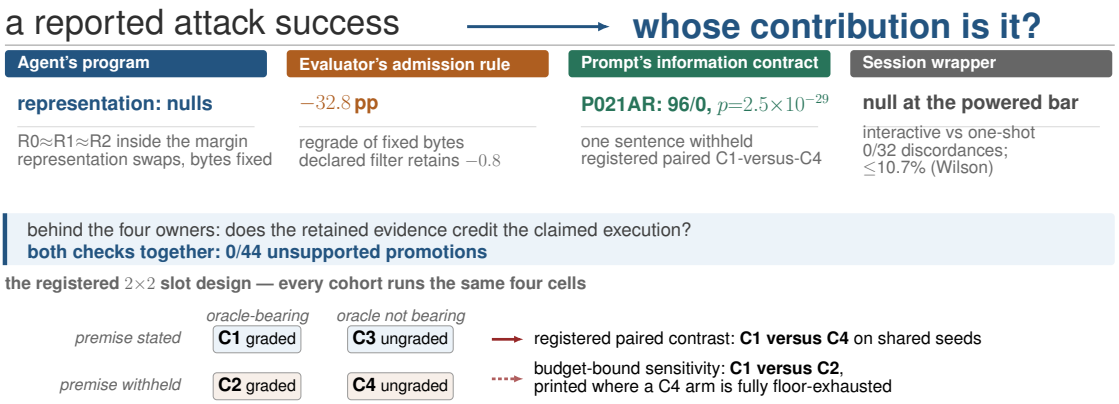


Figure 1. The ownership audit in one view. Frozen, one-owner-at-a-time interventions decompose a reported attack success (top): each card names an owner, its measured share, and the intervention (Sections 4–6); behind them stands the evidence question with its own measured answer (Section 9.1). The registered 2×2 slot design (bottom) keys every cohort: C1 states the zero-reference premise, C4 withholds it, and the registered contrast is C1-versus-C4 on shared seeds.

Registered external limits bound all of it (Section 7): in the pinned census 4/14 crypto tasks qualify. The repaired regrade carries 31/37, no correct program lost, no contrast moved (F1 FALSE). A second census finds 0/40 crypto tasks (CVE-Bench). A third meets the quorum (7/16 NYU CTF Bench candidates) and returns a non-degenerate measured null (261/282 admitted, 0 oracle-correct, F1 FALSE on all three labels).

The fourth question is execution attribution (RQ4; experiment code E5): on a held-out constructed case set, join-only checks incorrectly promote 16/44 attack-family cases, signature-only 28/44, their combination 0/44 — policy tests under trusted-role assumptions, not deployed attacks. A fifth follows the number outward: whether a credited execution carries security consequence (RQ5; §8).

1.1. Research Questions

- **RQ1 (admission; §4):** with program bytes and outcomes fixed, how much of the contrast does the admission rule control, and what mechanism triggers it?
- **RQ2 (information contract; §5):** which prompt disclosures move construction, and does the answer survive further labels, modern-AEAD fixtures on CSPRNG instances, and a negative control?
- **RQ3 (wrapper; §6):** does an interactive wrapper change construction relative to one-shot, and through which channel?
- **RQ4 (attribution; §9.1):** which authenticated bindings credit the claimed execution?
- **RQ5 (consequence; §8):** does a credited execution chain carry real security consequence, and does the effect transfer within-family?

Table 4 (Section 10) is the takeaway.

1.2. Contributions

The estimand. In a hash-frozen cohort, the *ownership share* of slot S is a paired contrast on shared seeds, $\Delta_S = P(\text{accept} \mid \text{slot } S \text{ varied}) - P(\text{accept} \mid \text{slot } S \text{ held})$, reported as an exact-binomial paired discordance with its interval — never a capability estimate, never a pooled rate: each cohort's Δ stands on its own population. The shares are not additive components of a reported number; the audit answers *which intervention moves the number*, not *what fraction of the number belongs to whom*. The interventions are not confined to rules of the authors' own making: the frozen records regrade under the published graders of three external benchmark teams, which agree with one another to the program. **Contributions: an estimand (each owner's share of a reported success), an instrument that measures it (hash-frozen, one-owner-at-a-time interventions), and a standard the measurement implies — the YHZ floor, five preconditions any future measurement can cite and comply with (Section 13).** The five research questions are one decomposition of that number, not five studies: three frozen interventions on its own owners, the evidence contract that credits the number to an execution, and the consequence that makes the credit matter. The frozen-counterfactual method moves one owner at a time, and the same instrument reports each owner's share:

(1) **Admission, with a tested mechanism:** regrading fixed bytes moves the contrast $-0.8 \rightarrow -32.8$; cleanup returns $+3.9$; 100/101 runtime traces show the unused-import trigger executes.

(2) **Information contract, with a cross-fixture null and a cross-label transport:** the cell collapses only under the named-with-unstated-premise conjunction in the original cohort (premise stated: C1 4/4, C3 4/4; premise withheld: C4 0/4, C2 4/4 at 6 vs 4 submissions; cell key: Figure 1).
Replications. The powered replication confirms the premise marginal and refutes the derive frame (C2 0/8). The n=16 cohort resolves the exact-test floor: paired C1-vs-C4 10/0, two-sided $p = 0.001953$. The second-label replication transports the premise share (paired 8/0, $p = 0.0078$). A fresh-seed n=16 cohort then reproduces the primary statistic exactly — paired 10/0, $p = 0.001953$; premise 32/32 vs 12/32 — registered as cross-vendor and withdrawn to same-label at runtime (Section 5).
The fixtures bracket the claim. A second hidden-premise family (AES-CTR) reads 0/16 in both withheld arms, and an observable-premise control (ECDSA) returns the pre-registered null (32/32 withheld). Three modern-AEAD fixtures (AES-GCM, ChaCha20-Poly1305, AES-GCM-SIV) repeat the

collapse on OS-CSPRNG-sampled instances, the last by design accepting nothing. The identical cell succeeds 4/4 on RC4 — the hash-frozen honest null (C4 4/4).

(3) **Wrapper, a null resolved at the powered size:** at ten pairs acceptance is indistinguishable (two-sided 0.65) at a cohort size the registered resolution shows cannot bound either direction; the pre-registered 32-pair cohort resolves the question at its registered bar (0/32 discordances in each direction; the advantage bounded at 10.7%, Wilson 95%); the only realized channel is the revisions clause (2 of 7).

(4) **External boundary, a non-degenerate measured null:** the first two censuses stop below quorum (4/14; 0/40); a third meets it (7/16) and its 168-session, three-label panel admits 261/282 programs — the repaired read boundary no longer blocks — but produces no oracle-correct program, so the registered contrast is a measured null (F1 FALSE on all three labels; F2 TRUE only vacuously at zero correct programs; F3 PASSED). A post-hoc decomposition attributes the null to named owners (Section 7). A registered frontier repanel then produced the record's first external solves — `lowe` byte-exactly on all three labels (12/12) — and measured the contract stack that left the 4-of-7 bar unmeetable in-panel. The honest reading is *unestablished at the registered bar for a capability reason* — no oracle-correct program existed for the contrast to move — not a judgment on the evaluated benchmarks, and not the earlier structural degeneracy (Appendix D.3).

(5) **Attribution, an evaluated contract:** execution credit requires authorized, cross-consistent records; ablations, bounded mutation testing, and an in-toto implementation check evaluate its behavior.

How to read this paper. Sections 4–6 move one owner at a time, each closing with its registered answer; Section 7 records the external boundary; Section 13 distills the five preconditions; Appendix A carries the registered session inventory (33 cohorts, 3,562 retained sessions, Table 5).

2. Related Work

Cryptographic capability of agents. The capability numbers exist; their ownership is unmeasured. The prior YHZ benchmark [1] reported a 40.9-point metadata-aided vs. blind gap and a 92.9% vs. 41.3% agentic vs. tool-only contrast on multi-tier LLM cipher identification — the reported numbers this audit takes as its target; the YHZ audit protocol introduced here is a distinct instrument. AICrypto [2], CryptanalysisBench [3], FirmAgent [4], APPATCH [5], CyberGym [6], AutoPT [7], and AutoPenBench [8] evaluate or exploit cryptographic and vulnerability tasks — direct capability comparators (per-system notes: Appendix E).

Our RQ2 departs from the prompt-sensitivity tradition [43], [44] in three ways: the intervention is matched (observations, target, seeds, budget, and interface stay identical while exactly one *semantic* prompt slot rotates); a pre-specified and hash-frozen replication on a second fixture returned an honest null, so the share is published with its boundary; and the ownership frame turns sensitivity into an accounted share. YHZ's own gap showed that information asymmetry moves measured capability but neither isolates the slot nor freezes the intervention; ABC [25], AgentSuite [26], and the Holistic Agent Leaderboard [52] localize the effect only at the family level. YHZ re-scores fixed outputs under a pre-registered intervention built to measure a named owner's share (Sections 4–5) — to our knowledge the first re-run of its own frozen instrument end to end, from the fixed-bytes regrade with output-preserving cleanup and runtime tracing (Section 4) to the matched 2×2 slot rotation (Section 5). Prior audits inspect or re-score fixed artifacts; this instrument intervenes, owner by owner, under a frozen contract.

Two adjacent literatures frame the reading: the *insufficient-information* line (models answer past missing evidence rather than abstain [47], [48]) and the *reasoning-budget* line (overthinking correlates with failure [49], [50]; test-time compute can be forced [51]). The withheld-premise arm instantiates the first pattern in an agentic security task with a paired design; the paper's floor-exhaustion mechanism (Section 5, P026) differs from both, indexing the deliberation excess to *withheld task information* and measuring it dose-responsively against a registered output cap. Prompt *injection* [45] mirrors the minus-direction withholding measured here, and the admission-share finding instantiates specification gaming's measurement problem [46].

Attack mechanisms. Cache-timing attacks on AES [9], [10], [11], [12], [13] and remote timing attacks [14], [15] establish the side-channel primitives we probe; protocol-failure attacks (FREAK [16], Logjam [17], POODLE [18]) with hybrid key-exchange design and deployment [19], [20], [21], [22], [23] define our target set's weaknesses. We introduce no new attacks: we measure whether agents execute the known classes.

Trajectories and evaluation integrity. CTF-ABACUS [24] provides trace-level provenance; ABC [25] audits task setup and reward design, including re-scoring fixed agent artifacts; AgentSuite [26] distinguishes user, environment, ground-truth, and evaluation defects; BenchJack [27] audits benchmark hackability; AgentRewardBench [28] compares automatic trajectory evaluations with expert judgments; PROV-AGENT [29] tracks workflow provenance. The Cybench and CVE-Bench enumerations test the hash-frozen portability of the mechanism against pinned snapshots — a different activity from benchmark-quality audits, on which we make no judgment.

Execution boundaries and attribution. AgentDojo [30], IsolateGPT [31], and SAGA [32] address execution isolation; ToolHijacker [33] shows tool-selection interfaces are themselves attack surfaces; ORTHRUS [34] illustrates making attack evidence attributable; Cloak, Honey, Trap [35] shows target-side deception can disrupt offensive agents. Our attribution experiments test which authenticated, mutually consistent records support an execution claim; the policy reuses existing provenance mechanisms, contributing its operational specification and measured behavior.

Public reconstruction evidence. A predeclared nine-field rubric applied to six releases [37], [39], [40], [41], [36], [42] examines the public evidence available for reconstructing a published outcome and yields a per-release census matrix (Appendix F.1); the fields it finds absent elsewhere are what this paper's release declares (Section 10).

3. Task Contracts, Threat Model, and Experimental Design

An evaluated attack result is a stack of separable decisions, and each decision has its own owner: whether the candidate satisfies the cryptographic predicate, whether the

program meets the declared admission rules, and whether the evidence supports crediting a particular execution (Table 1).

TABLE 1. CORE TERMS. CORRECTNESS, ADMISSION, AND ATTRIBUTION CONCERN DISTINCT OBJECTS.

Term	Meaning in this paper
Task contract	Declared observations, interface, budget, and success condition.
Information contract	Prompt-level disclosure frame: family naming and premise values (e.g., a zero reference).
Cohort	One registered population of sessions under a single frozen protocol on one model label.
Cell, arm, fixture	One prompt-slot condition (C1–C4); an arm is a cell's sessions within one cohort; the fixture is the construction instances are drawn from.
Attack program / candidate	Generated executable code and the value it submits to the task-specific check.
Admission policy	Source/runtime rules determining whether a program may execute.
Oracle / predicate	Trusted checker / relation the candidate must satisfy.
Verified outcome	Candidate accepted by oracle and predicate.
Attribution	Evidence binding an outcome to the claimed execution.
Ownership share	Paired contrast Δ_S on shared seeds; never a capability estimate or pooled rate.
Slot	One declared task-contract variable the audit intervenes on while everything else stays frozen.
Registered falsifier	The pre-stated observation that would refute a cohort family's claim; verdicts print per cohort — TRUE = the pre-stated falsifying observation obtained, FALSE = it did not; for null-design cohorts the registered expectation is itself the null, and the verdict reads against that expectation.

3.1. Threat Model and Trust Assumptions

Recovery task and measurement failure. The agent receives public observations, a disclosed weakness class, and the task's tools and budget, and must produce a predicate-satisfying program; reserved secrets and grading authority belong to the evaluator. An evaluator can misstate capability without an adversary (an undeclared source restriction rejecting a cryptographically correct program), and we test that mismatch separately from cryptographic failure.

Attribution adversary. The claimant and claimant-supplied evidence are untrusted; the constructed cases exercise fabricated claims, altered records, cross-execution splicing, and replay. The policy trusts the configured observer and reference roles, their signing keys, reference predicate, and identity and replay rules. It requires that accepted attribution refer to the execution and output those roles witnessed (Section 9.1); compromised trusted roles, arbitrary-host soundness, and exclusive secret-acquisition paths are outside the guarantee. Three decision-level threats motivate the instrument's design: benchmark inflation, attribution failure, and evaluator capture (the grading rule owning an unmeasured share of a published result). Each has a measured instance in this paper (benchmark inflation and evaluator capture in Sections 4 and 7; attribution failure in Section 9.1). The deployment consequence is direct: a SecOps pipeline gating agent deployment on a published rate inherits that rate's owners — the mis-owned number reaches production as a false sense of security.

3.2. Operational Setting and Contract Model

A task is described by opportunity O, recovery transformation T, attack code A, information/interface boundary B, verifier V, and predicate P; the contract fixes observations, imports, authority, budget, and predicate before generating A. The *Attack–Evaluation Oracle Boundary* separates attacker-visible observations from evaluator truth; a declared boundary is an assumption to check, not evidence of host isolation. "Attack success rate" is the genre's term for these recovery outcomes; the threat model here is measurement and attribution failure — no deployed system is attacked (Section 5).

3.3. Experimental Identification and Evidence

Evidence tiers. V denotes independently verified candidate/contrast evidence with the stated identity and trace; R, retained records missing a reconstruction link; E, estimates rather than executions. The release audit is A-tier; a V-tier candidate does not make every timing or attribution claim V-tier. Inferential status is a separate label from provenance tier: a result is confirmatory only when its registered analysis is frozen before the first model call (the C1 cohort's declared Holm family is the strictest instance; Section 11); analyses outside a freeze print exploratory, post-hoc, or descriptive at their sites, and qualification and preservation checks report instrument execution, not capability. The strict provenance conjunction H1 — not the H1-R3 study; the labels are unrelated — asks whether each session retains an immutable hosted-model image digest: V4 satisfies it in 0 of 12 sessions, a local digest-pinned OCI control in 3 of 3 (a local identity link only).

3.4. Capture-First Harness and Qualification

With the harness qualified, Section 4 intervenes on the first owner: the evaluator's admission rule.

4. Separating Construction from Admission

RQ1 asks what separates an accepted program from a rejected one when the program bytes do not change. Three interventions answer it: H1-R3 changes the representation with task and evaluator fixed; counterfactual regrading holds the programs fixed while changing admission; import cleanup tests the mechanism. The reduced-information diagnostic shows the information frame also moves outcomes; Section 5 isolates the sentence, and Section 8 (RQ5) the target-effect claims.

4.1. Matched Construction Study (H1-R3)

A pre-specified, hash-frozen matched cohort isolates representation by fixing the task, target, route, parser, tools, feedback, budget, executor, and reference oracle on a single `claude-sonnet-4-6` label. It compares a neutral representation (R0) with two alternatives on XOR keystream-reuse and ECDSA nonce-reuse tasks at three complexity levels. L1–L3 increase observation processing, not cryptanalytic depth (level composition: Appendix E). R0 uses compact JSON and hexadecimal observations; R1 changes key order and whitespace; R2 uses base64. The 852 scheduled sessions comprise 768 representation trials, 64 repeat-prompt controls (R4), and 20 abstention controls; none enlarges the primary comparison. Protocol and freeze trail (Appendix B).

At the powered L3 level the three representations succeed at nearly identical rates: 125/128 (R0), 126/128 (R1), and 124/128 (R2) sessions. The paired effects are 0.008 and -0.008 with 95% intervals $[-0.023, 0.039]$ and

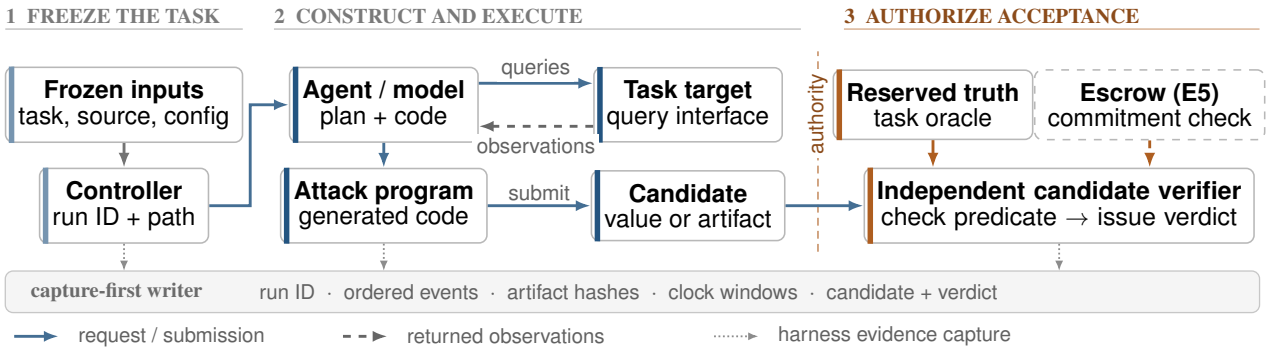


Figure 2. The agent constructs code and interacts with the target; the independent verifier checks its candidate against reserved truth; the controller freezes inputs and reserves run identity (dotted: harness-owned capture-first writer; dashed: optional E5 escrow). Regions identify trusted responsibilities under Section 3.1's assumptions; they do not prove host isolation.

$[-0.047, 0.031]$, Holm-adjusted exact McNemar $p = 1.000$ for both, within the pre-specified ± 0.10 margin. The equivalence conclusion rests on the intervals against the margin and the companion exact bound $(1 - 0.025^{1/N})$ (Appendix B), not the vacuous McNemar value at these discordance counts: no large pooled representation effect under this contract — the near-ceiling baseline leaves little headroom and bounds no negative effect. The lower-level R2 contrasts cross the margin and are inconclusive (Section 11). Companion bounds (Appendix B).

4.2. Same-Program Admission Counterfactual

This intervention changes the evaluator decision on existing outputs; it generates no new programs and is not an independent replication. *The first owner named is our own harness:* we wrote the earlier import filter and never declared it in the task contract; regrading fixed bytes under it is a self-audit of undeclared rule variance. The self-audit generalizes in method, not magnitude, and only where evaluators regrade their own frozen outputs; the owners themselves are structural positions — any pipeline that filters programs, states an information contract, wraps sessions, and binds evidence carries the same four slots whether or not anyone measures them. The earlier filter would reject 6.2%–47.7% of the retained programs across arms; holding programs fixed isolates the admission decision, and a source rejection precedes execution, counted separately from cryptographic failure. The complete cohort contains 801 verified-exploit successes, 20 correct abstentions, 28 cryptographic-predicate failures, 2 execution failures, and 1 source-qualification failure — 852 sessions in all.

For instance i and arm r, let a_{ir} be the retained program, K_{ir} its reference-checked success, and $A_p(a_{ir})$ its admission under rule p. The scored contrast on the fixed set of $n = 128$ L3 pairs is

$$\widehat{\tau}_p = \frac{1}{n} \sum_i \left[A_p(a_{i,R2}) K_{i,R2} - A_p(a_{i,R0}) K_{i,R0} \right].$$

The program bytes and K remain fixed. Changing only A_p moves this contrast from -0.8 to -32.8 percentage points: 50 instances shift against R2, 9 for it, 69 do not change. The regrading is exploratory on a finite cohort; it does not revise the pre-specified L3 analysis under the declared rule. This $\widehat{\tau}_p$ is the admission arm of the ownership share Δ_S defined in Section 1. The rule actually in force admits exactly the seventeen declared modules, rejecting none of the 768 representation programs and leaving that contrast at -0.8: the 32 points measure an admission artifact of a rule narrower than the task contract, not a difference in capability (the undeclared-filter record: Appendix E).

Most of the change traces to one module: of the 852 sessions, 205 import a module the earlier rule denies — in every case a module the prompt permits; denying `hashlib` alone accounts for -28.9 of the -32.0 points, and no other single-module narrowing moves the contrast by more than 3.1 points (per-arm import-rate detail: Appendix E).

Output-preserving unused-import cleanup eliminates the large negative contrast: of the 128 cleaned derivatives (all reproducing their original output), 104 qualify and 98 are cryptographically correct. The filtered L3 contrast changes from -32.8 to $+3.9$ points, against -0.8 under the declared contract; the 4.7-point residue above the declared-rule value is unexplained by the retained records and recorded as such. The trigger is executed code, not a classifier artifact: of the 101 denied-module sessions whose runtime traces retained the import reference, the imported object load was observed in 100 of 101 executions; the exception's trace records no observed load (per-session trace census: Appendix E).

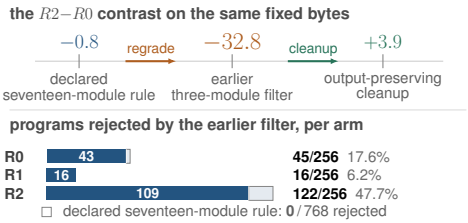


Figure 3. An undeclared evaluator rule, not a weaker agent: import rejections under an earlier three-module filter narrower than the task prompt — 45/256 R0, 16/256 R1, 122/256 R2, totaling 183 of 768 representation programs; dark segments are cryptographically accepted (43, 16, 109), gray the rest (2, 0, 13); the declared seventeen-module rule rejects none. Program bytes stay fixed: on 128 matched L3 instances the filter changes the R2–R0 contrast from -0.8 to -32.8 points; the declared rule retains -0.8.

Answer to RQ1. Under the declared contract, the two pooled primary L3 intervals lie within the frozen margin; the mismatched filter creates a large negative R2 contrast, and unused-import cleanup removes it. The ECDSA diagnostic separately identifies a domain-parameter defect; its arm-asymmetric prevalence is a constructed, non-confirmatory diagnostic, not a representation effect. These findings distinguish implementation failure from evaluator rejection; their prevalence beyond this cohort is unmeasured (the pre-specified external attempt: Section 7).

5. The Information Contract: What a Single Sentence Owns

The result in one sentence: with the zero-reference premise stated, the stated arm accepts every evaluable session in every Claude- and GPT-family cohort on a premise-bearing fixture; with it withheld, no withheld seed in any of the sixteen registered family cohorts beats its stated counterpart. Three printed exceptions flank this: the AES-GCM-SIV control accepts nothing in any cell by design, Moonshot's stated arm is not saturated by measurement, and the n=32 scale-up's damaged first pass is closed by its registered 64/64 rerun. Outside the family, the kimi reruns hold the three wins disclosed below. This section measures what that sentence owns.

Twenty-four pre-specified and hash-frozen factorials hold an entire session fixed and intervene on exactly two prompt slots: whether the family and weakness are named, and whether the reference premise is stated or merely posited with unstated content — the premise being the fixture's declared convention about its reference (for ChaCha20, the statement that the reference encrypts the all-zero message). The same four seeds rotate through the four cells in a Latin square per cohort, and the freeze precedes the first model call. Each cohort runs one provider-attested model label: four primary-label cohorts plus three same-label controls on `claude-sonnet-4-6` (P011, saturated RC4; P012, observable-premise ECDSA; P013, the AES-CTR boundary run), three further labels, and two on `kimi-k3` (per-label roster in Section 11).

Model labels and routes. Every cohort label in Sections 4–7 — `claude-sonnet-4-6`, `gpt-6.1-sol`, `gpt-5.6-sol`, `gpt-6-astra`, `kimi-k3`, and the three NYU-panel labels of Section 7 — is a provider-attested endpoint through a commercial LLM gateway: a label name identifies the route as attested at call time, not a vendor product. No route is digest-pinned; sessions bind to labels by the post-run fingerprint audit (every raw response checked against the attested name; the one drifted cohort withdrawn to same-label, Section 11). The gateway's attested route strings carry a tier prefix the paper omits; they live in the released run records (routes, attestations, substitution boundaries: Appendix D).

Each cohort is registered individually; separate populations, never pooled for inference (the shared rule: Section 11). The four cells (C1 premise stated, oracle-bearing; C2 withheld, oracle-bearing; C3 stated, no oracle; C4 withheld, no oracle; Figure 1, bottom) run through every cohort; naming is not a free third factor (C1/C4 name the family and weakness, C2/C3 leave them unnamed — a withheld arm cannot name a premise it cannot see, so the naming difference tracks the premise arm by construction): the registered primary C1-versus-C4 holds naming fixed and moves the premise and the oracle channel together, while its budget-bound C1-versus-C2 sensitivity holds the oracle channel fixed and moves the premise with the naming. The design is repeated measurement, not accumulation: every cohort re-runs one frozen protocol on fresh seeds under a resolution analysis that predates every call. The roster, per-cell n, and per-cohort records sit in Appendix D and Appendix E; Figure 4 plots the withheld-arm totals. P010 record (Appendix D).

Reading the statistics. The primary statistic is the paired C1-versus-C4 discordance: shared-seed disagreements, read C1-over-C4, exact binomial, Holm-corrected at FWER 0.05 across the sixteen registered tests. Premise marginals and cross-cohort counts are descriptives; Wilson and Clopper–Pearson intervals are a near-zero rate's honest range.

On ChaCha20 (P010, 16/16 sessions, 0 provider failures), the cell collapses under exactly one slot combination: name stated, premise withheld. The paired C1-versus-C4 contrast reads 4/0 ($p = 0.125$), the premise marginal 8/8 vs 4/8, naming inert; every stated-premise acceptance is first-turn. The registered falsifiers read F1 FALSE (C4 matching C1 would weaken the premise reading), F2 FALSE (a C3 failure beside a C1 success would hand the outcome to the oracle), F3 TRUE (C2 succeeds — with oracle access the premise is unnecessary); full falsifier and failure-stage record: Appendix E.

The chain then escalates by design, each step registered against the previous one's critique. The powered same-label replication (P010P) reads paired 5/0; the n=16 cohort (P010N16) 10/0; the second and third labels transport the premise share (8/0 each). The fourth label is the printed boundary condition. The fifth label (Moonshot K3) brackets the stated arm — null and collapse both clean, the degree cohort-relative; at n=16 apiece the two cohorts' contrasts do not separate statistically, so the bracketing is descriptive rather than inferential. A fresh-seed n=16 cohort (P010X1) repeats the paired contrast at 10/0 on new seeds (full records: Appendices D and E).

The four cells, and where the withheld arm's failures come from. Each cohort runs the same four cells (Figure 1, bottom). Across the wave, every withheld-arm non-acceptance is one of two measured kinds: a *text* submission that fails the predicate, or a *floor-exhausted* session — the model's hidden deliberation outlasts the registered output floor and the route returns an empty text block (68/68 signature agreement; P026 confirms the mechanism at the raised floor, 4200→8000 tokens: Appendix D). Floor-exhausted sessions are a measurement-layer outcome — never acceptances, never transport loss — reported per cohort in the released supplement's record S30 (protocol: Appendix E).

The registered scale-up completed: P011-L3-N32 (third label, per-cell n=32, 128 sessions, hash-frozen before its first call) first read paired 17/0 with fifteen ties (two-sided $p = 1.5 \times 10^{-5}$), its stated arm then below saturation (C1 27/32, C3 30/32) and its withheld arm 0/36 evaluable (Wilson 95% CI $[0\%, 9.6\%]$). Its clean rerun on the repaired transport removes the gap: 32/0 ($p = 4.7\times10^{-10}$, Table 2; stated 64/64, withheld 0/61 evaluable). P011-L3-N32 record (Appendix D).

The modern-AEAD wave (P014–P024, Table 2). This registered wave repeats the factorial on current constructions and repairs the hand-declared-seed sampling frame: P018–P020's instances are drawn from the OS CSPRNG before registration (24, 24, and 8, hash-registered at draw time); the replication unit becomes a random instance, not a fixed one. Powered paired discordances run 12/0 to 96/0 (Table 2; the two floor-exhausted-only arms report by their budget-bound sensitivities). P021AR (repairs P021) reads paired C1-versus-C4 96/0 (p=2.5×10^{-29}; the budget-bound C1-vs-C2 contrast reads 96/96 at the same p), stated arm 192/192 evaluable, withheld arm 0/141 evaluable with 51 further withheld-unnamed sessions floor-exhausted (measured at the floor, not lost); under censoring the estimand is the paired contrast on evaluable pairs, printed beside the all-registered-session sensitivity that every enrolled withheld session non-accepted at its floor, never an imputed rate.

TABLE 2. THE REGISTERED SCALE-UP AND THE MODERN-AEAD WAVE (THE INSTANCE IS THE UNIT; SEPARATE POPULATIONS, NEVER POOLED). EVERY COHORT IS READ FROM ITS COMPLETED REGISTRATION: THE 2026-11-03 TRANSPORT REPAIR RE-RAN EACH DAMAGED FIRST PASS ON THE REPAIRED ROUTE (MARKED "RERUN"), AND EVERY FIRST-PASS LOSS IS ATTRIBUTED BY THE RECEIPTED CENSUS. P018 IS THE EXCEPTION ITS ROW STATES: ITS RECOVERY RERUN IS LABEL-SPLIT, SO ITS ROW READS THE REGISTERED-LABEL FIRST PASS AND DISCLOSES BOTH RECOVERY HALVES. PAIRED = C1-VERSUS-C4 DISCORDANCE UNDER THE REGISTERED EVALUABLE-PAIRS RULE (A FLOOR-EXHAUSTED SESSION IS A SEPARATE TERMINAL OUTCOME, NOT A SCORED NON-ACCEPTANCE); STATED VS WITHHELD = PREMISE MARGINAL ACCEPTED/EVALUABLE; WHERE A LABEL'S C4 ARM IS FULLY FLOOR-EXHAUSTED, THE BUDGET-BOUND C1-VS-C2 SENSITIVITY IS PRINTED INSTEAD AND MARKED. COHORTS RUN ON `GPT-6-ASTRA` EXCEPT WHERE A ROW STATES OTHERWISE; THE P016 ARMS CARRY THEIR OWN MODEL LABELS, AND THE P022/P024 ARMS LIKEWISE. THE ORIGINAL 2×2 (P010, P011) APPEARS IN APPENDIX D.

Cohort	Fixture	Paired	Stated vs withheld
P011-L3-N32 (scale-up, rerun)	ChaCha20	32/0 (4.7×10^{-10})	64/64 vs 0/61
P014 (seeds 401–416, rerun)	AES-GCM	13/0 (2.4×10^{-4})	32/32 vs 0/14
P016 (`gpt-6.1-sol` arm, rerun)	AES-GCM	12/0 (4.9×10^{-4})	31/31 vs 0/12
P016 (`gpt-5.6-sol` arm, rerun)	AES-GCM	no evaluable C4 (floor-exhausted); C1-vs-C2 15/0, $p = 6.1\times10^{-5}$	
P017 (floor 4200, rerun)	AES-GCM	16/0 (3.1×10^{-5})	32/32 vs 0/25
P018 (CSPRNG, 24)	AES-GCM	22/0 (4.8×10^{-7}); recovery 12/0 + 12/0 (4.9×10^{-4} each, label-split)	46/46 vs 0/37
P019 (CSPRNG, 24, rerun)	ChaCha20-Poly	24/0 (1.2×10^{-7})	48/48 vs 0/34
P020 (CSPRNG, 8)	AES-GCM-SIV	negative control: no cell accepts	
P021AR (CSPRNG, 96; repairs P021)	AES-GCM	96/0 (2.5×10^{-29})	192/192 vs 0/141
P022a (CSPRNG, 24; `gpt-6.1-sol`, rerun)	AES-GCM	24/0 (1.2×10^{-7})	48/48 vs 0/37
P022b (CSPRNG, 24; `gpt-5.6-sol`, rerun)	AES-GCM	no evaluable C4 (floor-exhausted); C1-vs-C2 24/0, $p = 1.2\times10^{-7}$	
P023 (CSPRNG, 96; fresh seeds)	AES-GCM	95/0 (5.0×10^{-29})	191/191 vs 0/141
P024a (24; `gpt-6.1-sol`; 8400)	AES-GCM	24/0 (1.2×10^{-7})	48/48 vs 0/48
P024b (24; `gpt-5.6-sol`; 8400, rerun)	AES-GCM	1/0 (1.0; 1 evaluable C4; 46/48 floor-exhausted)	48/48 vs 0/2

Three recovery passes (P018's remaining pairs, P023's, P026's) completed after the registered label's quota exhaustion on the author-directed substitute label `gpt-5.6-sol`; every number keeps the two label groups separate. P023's recovery pass, split at its recorded boundary, re-read 340 sessions on the registered label (C1-versus-C4 83/0, p=2.1×10^{-25}) and completed the remaining 44 sessions on the substitute label after the quota ban (C1-versus-C4 11/0, p=9.8×10^{-4}). One registered cohort carries the same one-way direction on two labels; no number mixes them.

P024a closes the truncation reading: *withholding the premise does not stop the model answering — it makes every answer wrong.* P024b is the printed boundary condition: consistent, under-powered at every coding. AES-GCM-SIV accepts nothing in any cell: the registered negative control.

Two defects travel with the wave, both disclosed: attrition concentrates in withheld cells and P018's recovery rerun is label-split (full per-cohort records: Appendix E).

Family-level correction over the chain's primary statistics. Inference is anchored to the registered family, not to any single cohort: the 16 evaluable paired C1-versus-C4 registrations completed after the 2026-11-03 transport repair whose primary statistic is informative (membership enumeration and exclusion rule: Appendix E). *The family at one glance.* Table 3 carries the sixteen members' registered statistics. Every cohort that stood alone at α=0.05 still clears the correction; the one that did not — P010P's $5/0$ borderline, Holm 0.0625 in the m=16 family — remains non-significant and prints as its own bound, never absorbed into a family claim. A sensitivity Holm over the full registered set holding evaluable primary statistics — the sixteen members plus the disclosed P024b boundary registration (seventeen tests; the P010L4 rerun holds no evaluable pair and registers no statistic) — changes no verdict: all fifteen survivors clear the tightened step-down (tightest margin 0.00781 vs. 0.0125 at position fourteen), and the $5/0$ borderline and the degenerate boundary statistic stay non-significant. Two boundary conditions sit outside the family under the same strict rule, disclosed rather than absorbed (P024b's completed registration, consistent but under-powered at every coding; the P010L4 rerun holds no evaluable pair and its registered statistic is not evaluable; Appendix E).

What is stable and what is not. The stable endpoint of this instrument is the *one-way discordance*: across every family member and every first-pass rendering printed in this paper, no $C4$ seed beats its $C1$ counterpart, so the paired contrast is one-way (*one-directional*) wherever evaluable. The one exception class is printed in full: the kimi label's clean reruns, outside the family, hold three withheld-direction wins across their 16 pairs on a label whose stated arm is itself non-deterministic (full record: Appendix E). The *degree* of the withheld-premise arm is cohort-relative, never a population parameter (Wilson 95% CIs per cohort; Figure 4); no row is exchangeable.

TABLE 3. THE REGISTERED 16-MEMBER FAMILY: EVERY PAIRED CONTRAST IS ONE-WAY. PAIRED = C1-VERSUS-C4 DISCORDANCE UNDER THE REGISTERED EVALUABLE-PAIRS RULE; p = EXACT TWO-SIDED BINOMIAL; HOLM = STEP-DOWN ADJUSTMENT AT FWER 0.05; CP = CLOPPER–PEARSON TWO-SIDED 95% LOWER BOUND ON THE ONE-DIRECTIONAL DISCORDANCE PROBABILITY. REGENERATED FROM THE RAW PER-SEED RECORDS BY ONE SCRIPT; THREE SIGNIFICANT FIGURES, ONE ROUNDING RULE.

Cohort	Fixture	Paired	p	Holm p	CP
P021AR	AES-GCM	96/0	2.52×10^{-29}	4.04×10^{-28}	≥ 0.96
P023	AES-GCM	95/0	5.05×10^{-29}	7.57×10^{-28}	≥ 0.96
P011-L3-N32	ChaCha20	32/0	4.66×10^{-10}	6.52×10^{-9}	≥ 0.89
P019	ChaCha20-Poly	24/0	1.19×10^{-7}	1.55×10^{-6}	≥ 0.86
P022a	AES-GCM	24/0	1.19×10^{-7}	1.55×10^{-6}	≥ 0.86
P024a	AES-GCM	24/0	1.19×10^{-7}	1.55×10^{-6}	≥ 0.86
P018	AES-GCM	22/0	4.77×10^{-7}	4.77×10^{-6}	≥ 0.85
P013	AES-CTR	16/0	3.05×10^{-5}	2.75×10^{-4}	≥ 0.79
P017	AES-GCM	16/0	3.05×10^{-5}	2.75×10^{-4}	≥ 0.79
P014	AES-GCM	13/0	2.44×10^{-4}	0.00171	≥ 0.75
P016 (`gpt-6.1-sol`)	AES-GCM	12/0	4.88×10^{-4}	0.00293	≥ 0.74
P010N16	ChaCha20	10/0	0.00195	0.00977	≥ 0.69
P010X1	ChaCha20	10/0	0.00195	0.00977	≥ 0.69
P010L2	ChaCha20	8/0	0.00781	0.0234	≥ 0.63
P010L3	ChaCha20	8/0	0.00781	0.0234	≥ 0.63
P010P	ChaCha20	5/0	0.0625	0.0625	≥ 0.48

The second hidden-premise family (P013, AES-CTR, seeds 181–196, n=16) repeats the pair on a third fixture: with the premise stated, every stated cell succeeds first-turn (C1 and C3 16/16); withheld, every session fails even when the family is named (C2 and C4 0/16, 31 and 27 submissions). The ECDSA control's premise is redundant in the public data, so withholding it costs nothing: every cell completes 16/16, the pre-registered predicted null (both records in the supplement).P012 record (Appendix D).P013 record (Appendix D).

The submitted code carries the mechanism: stated-

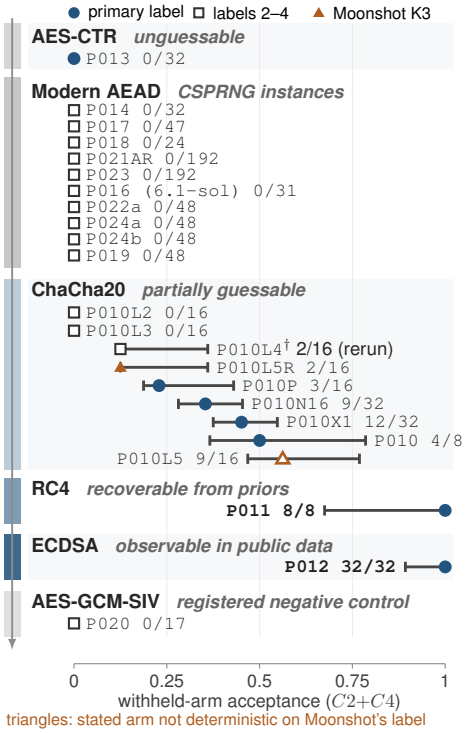


Figure 4. Withheld-premise success tracks how obtainable the hidden content is from non-prompt sources. Per-cohort registered withheld-arm totals ($C2{+}C4$; rows juxtapose cohorts, never pooling), Wilson 95% intervals; circles: primary-label cohorts, squares: labels 2–4, triangles: Moonshot's label (open: degraded route, filled: healthy route). The stated arm is 100% on every Claude and GPT-family label at per-cell $n \leq 16$; on Moonshot's it is not, and the GPT-family scale-up is the exception at $n{=}32$ (C1 27/32, C3 30/32). $*$: P010L4 is read under the registered non-acceptance rule. The fourth label's clean rerun measures its withheld arms (2/16 accepted, 14 floor-exhausted under the 2800-token floor).

premise programs build directly; derive-frame programs conjecture the zero-reference convention and pay in submissions; C4 programs chase the reference content itself — a known-plaintext reading, not an internal-state claim.

The structural replication on RC4 (P011) returns the anticipated honest null: C4, the cell that collapsed on ChaCha20, completes 4/4; both marginals sit at $p = 1.0$; the paired discordance is 0/0 and not evaluable (F1 TRUE, the honest empty premise). The code route recovers via an assumed zero reference, and a retained underdetermination witness shows the instance is not solvable from the public fields alone (full record: Appendix E). P011 record (Appendix D).

Read together, the twenty-four cohorts separate two evaluator interventions usually conflated: withholding a disclosure and framing a gap without supplying its resolution. The inferential headline is the pre-specified and hash-frozen paired C1-versus-C4 contrast: one-way in all sixteen family members (Table 3), fifteen Holm-significant at FWER 0.05, with the 5/0 borderline printed as its own bound.

Premise-marginal counts are descriptive: 56/56 vs 16/56 same-label, 16/16 vs 0/16 on each of the two other labels, 16/16 vs 2/16 on the fourth. The pooled Fisher treats all 112 sessions as independent, so it is not reported; the seed-stratified test below carries the inference. Naming does not gate while the premise is stated: C3 saturates in every Claude- and GPT-family cohort at per-cell $n{\leq}16$, with Moonshot's stated arm the second printed exception and the $n{=}32$ first pass the third. The derive frame collapses outside the original cohort. The one cross-cohort summary computed, a seed-stratified CMH (Cochran–Mantel–Haenszel) test at $1.5{\times}10^{-12}$, is a labeled post-hoc descriptive.

The freeze binds the instrument and protocol, not the model: the rerun measures exactly that, and the three-fixture observability axis bounds the share to premises about secret content (drift caveats, the post-hoc rerun, and the bounding argument in full: Appendix E).

Answer to RQ2. The premise slot owns the cell: the registered paired C1-versus-C4 contrast is one-way in all sixteen family members (fifteen Holm-significant at FWER 0.05; the 5/0 borderline printed as its own bound), naming is inert, the derive frame is refuted outside the discovery cohort, and the share is bounded to premises about secret content. The registered contrast varies the premise and oracle-bearing slots together; the premise-only reading is carried by the registered C1-versus-C2 sensitivities — printed for every floor-exhausted arm and for P021AR's 96/96, one-way wherever evaluable — and the C3-versus-C4 single-slot read is not separately powered at these cohort sizes and is not claimed. Section 6 moves the intervention from the prompt to the session.

6. Interactive versus One-Shot Construction: A Bounded Wrapper Null (H2)

Does letting the agent iterate rescue the number? RQ3 moves the intervention from the prompt to the session and asks it where a null can still be read. One CBC padding-oracle task, seeds, prompts, and a local, deterministic (non-timing) feasibility gate back two session wrappers: A_INTERACTIVE may submit, receive the fixed rejection feedback, and revise once within a turn-2 cap; B_ONESHOT submits once. The gate passes on every seed within the 4,096-query budget (frozen 1,842–2,238 queries, seeds 61–64; extension 1,633–2,679, seeds 65–74), consistent with the 2,021-query control. Two pre-specified and hash-frozen cohorts run this design, one provider-attested label each, analyzed separately, never pooled. Frozen cohort (Appendix D).

The powered cohort resolves the question at its bar. A pre-registered 32-pair cohort runs both arms on every seed under the registered paired bar (six one-directional discordances, $p = 0.03125$). It quadruples the extension on the same frozen fixture, wrappers, feedback, policy, gate, and turn flow (seeds 441–472; label `gpt-6.1-sol`, route-default temperature, an 8,400-token floor). All 64 sessions accept on the first turn: the paired discordance is 0/32 in each direction, bounding the interactive arm's advantage at 10.7% (Wilson 95%), and the turn-2 revision pathway is never entered (zero turn-1 rejections, using 1,384–2,562 oracle queries, none over budget). The route's content filter intermittently flags large-budget requests; a registered retry policy (at most six attempts on pre-output conditions, logged per call, identical for both arms) delivered every session. At frontier capability the session wrapper owns no share of this number: the registered null holds at its bar.

Answer to RQ3. At the ten-pair registered size the wrapper question was unresolved: acceptance is descriptively indistinguishable at ten pairs, a size at which the power analysis cannot bound either direction; the only realized channel then is the revision path. The powered cohort then resolves the question at its registered bar (32/32 pairs tied; the interactive arm's advantage bounded at 10.7%, Wilson

95%). The session wrapper's measured share is diagnosed post hoc as a *format-compliance effect*, not an information effect: the post-hoc re-adjudication recovers three of the four parse failures (post-hoc acceptance 8/10 against B's 7/10), the oracle-separation control bounds the information channel (0/10 with 256 probes, the information-absence signature), and the combined paired view is tied at 9/9 — A's programs were not worse; A's second turn was unreadable. The null's reach is the tasks a frontier model solves on the first turn: where the first turn fails, the revision pathway is the only measured channel (2 of 7), and whether interaction pays on tasks that genuinely require iteration stays registered-open (Section 11) — the open cell being reasoning-loop agents on multi-step planning tasks. The external repanel gives the pathway its first external exercise — 23 of 36 frontier sessions resolved at turns 2–4. Extension cohort (Appendix D).

7. External Boundary: the Panel, its Cohorts, and its Owners

Does the admission mechanism of Section 4.2 travel beyond our own harness? The external panel answers with three measured cohorts on two pinned benchmark archives, one census that closes a third source, and a fully decomposed floor; every layer is reported in its final measured state. One heading governs the reading: *no statistic in this section moves another team's published number* — the cohorts register where the instrument travels and where it stops, every boundary the panel hits is named by its owner: the transport, the answer normalization, the admission interface, the benchmark domain, and the models' capability floor — and what the section does establish is rule provenance: the answer-normalization owner instantiates under the published graders of three external teams, applied to our own frozen outputs.

The sources. A complete census of the pinned Cybench snapshot found 14 crypto-category tasks, 4 of which qualify under the pre-specified and hash-frozen rule (Makeshift, Dynastic, Primary Knowledge, Partial Tenacity; stdlib-only reference solvers reproduce all four pinned answers; decode-equality PASS 4/4). Nine are service-bound and one is excluded on solver requirements; With 4 against a quorum of 6 (the registered minimum, tied to the design's power statement), even admitting the excluded task gives $5 < 6$, so the Cybench slice carries no falsifier verdict, only the census. A second census closed the expanded-source question: all 40 CVE-Bench task targets are enumerated from the pinned archive and every one is a web-application CVE (WordPress plugins, CMS and PHP applications, Python services, one Rust service) — zero cryptographic-library targets, and every task is service-bound by construction (its metadata schema carries exactly 12 keys with no categories field), so no entry qualifies. The crypto slice of the pinned NYU CTF Bench release supplies the panel: 75 crypto rows, 16 offline candidates, 7 tasks that satisfy the full criterion-5 gate (quorum 6 met) — a registered selection rule that requires a frozen, offline, stdlib-reproducible oracle, so the external record speaks to these seven tasks, not NYU's crypto category at large. Census records (Appendix D). CVE-Bench census (Appendix D).

Cohort B — the NYU panel on the free-tier labels. The registered three-label panel (`gpt-6-luna`, `qwen-3.8-max`, `deepseek-v4.1-flash`) ran 168/168 terminal sessions on the seven NYU tasks: 12 provider-no-content exclusions (HTTP 429), 156 content sessions, 282 retained programs. No program is oracle-correct. The declared arm qualifies 261/282 programs (92.6%) and the stress-repaired arm 149/282 (52.8%): the admission interface is not the blocker on this cohort. With zero correct programs, the per-label representation contrast is 0 on every task with a denominator and undefined on the rest; F1 is FALSE on all three labels (0 movers; registered threshold 4 of 7; movement defined on 4, 5, and 4 tasks respectively), F2 TRUE (vacuous), F3 PASSED_preservation (282/282), and the cross-label rule is not met. The floor decomposes into named owners: *eleven* sessions' outputs print the correct `eps` flag (8 at final submission) — all rejected by the registered whole-stdout rule while the benchmark's own flag-in-output diagnostic reports flag found in all eleven, so the answer-normalization owner splits the near-solve across codings. A pre-registered regrade of the same frozen captured-output windows under the published graders of three external teams — NYU CTF Bench's native flag check, InterCode-CTF [38], and Cybench — scores 18 of the 282 retained programs (11 sessions) correct, all three agreeing with the permissive coding to the program; the declared admission arm admits all eighteen and the stress arm discards sixteen, so the admission contrast on correct programs exists only under the external graders. *The zero is a measured coding point, not an instrument boundary:* the instrument ran unmodified on the benchmark's own tasks, every owner it names on the home harness is named and measured here, and the three external graders agree with one another and with the instrument's permissive coding — what stops at zero is the strict whole-stdout rule, one declared point in the normalization owner's range. *A hundred and three* sessions' final program crashed, and the crash class decomposes fully at the stderr level with zero harness-caused members: 48 are one decode idiom (36 of them on the RT representation, where the prompt's decode note primes a base64 read of a payload that is plain bits — the decode note overrode the file's self-evident content, an authority-over-evidence failure of exactly the kind this paper measures; the other 12 are execution failures on payloads that are genuinely base64, the correct solve path failing mid-run), 54 are logic failures, and one is a hallucinated dependency on a wordlist file the task release does not ship. RT delivery verifies byte-exact in 84/84 prompts. The registered verdicts are unchanged; the decomposition replaces a flat zero with named owners. Amendment-3 record (Appendix D).

Cohort D — the published-protocol audit (M1). NYU CTF Bench's crypto rates are published at 0–5.77% across five models, and its release ships no per-run records — the published numbers cannot be audited from the release. The audit therefore mounts the published protocol itself on the same hash-frozen inputs, the census's 12,000-token ceiling treated as a declared variable — capability boundary or instrument boundary (full grid and amendments: the Cohort D record). Under the declared allowance 20 of 50 scored sessions solved and 4 of 10 task-level pairs, identical under all five evaluator rules — 30 sessions never acted at all, every turn spent in unshown reasoning, so six of ten pairs are *unmeasured rather than failed*. Two allowance amendments re-measured exactly those slots (32,000 tokens; 64,000 where truncation persisted): the scored set closes at *28 of 50* session-level and *8 of 10* task-level, identical across the five rules, with 19 ceiling-censored sessions and 17 zero-turn HTTP-504 slots reported transport-unmeasured, never as model

failure. The declared allowance under-states the record it can report: the zeros were instrument-bound, and every rate this paper quotes from that grid carries the allowance it was measured at. Published-protocol audit record (Appendix D).

The decomposed floor. Across the four cohorts the external record measures four owners, each on its own cohort and never pooled: the transport (empty-completion truncation on the facade route; zero on the responses route; and, in the audit, a per-call allowance that truncates reasoning before any action at 12,000 tokens and yields solves once raised), the answer normalization (statement-faithful `eps` output accepted by the benchmark's own diagnostic and rejected by the registered whole-stdout rule, in both cohorts that reach `eps`), the admission interface (the read-boundary clause rejecting correct programs on the HTB and frontier cohorts while admitting 92.6% of the free-tier cohort's programs), and the capability floor (zero correct programs on the free-tier labels; 35/36 on the frontier labels; zero sessions on Cybench's slice, whose qualified tasks the quorum cannot seat). The portability falsifier F1 reads FALSE everywhere it is evaluable and not-evaluable where the census shrinks its denominator; what the external record establishes is not that the mechanism fails to travel, but precisely which owner stops it each time — measured, named, and bounded to the grids that produced it. Frontier facade record (Appendix D).

8. From Recovery Programs to Security Consequences

RQ1 isolates an admission artifact. The last two questions step outside the number's owners and examine the result itself: whether the retained evidence credits the claimed execution (RQ4; Section 9.1), and whether a credited execution carries security consequence (RQ5). RQ5 asks: can the supplied operation execute, can an agent construct its program, which observations enable recovery, and does the target predicate establish a security consequence? Each study answers one question under its own contract; the sequence is an evidence map, not a single end-to-end campaign (construction spans and chain analysis: Appendix E).

8.1. Prospective V4 Cohort

The frozen prospective cohort (EXP-034) runs 12 cells once with no replacement (claude-sonnet-4-6, temperature 0.2, 4096-token budget): eight sessions reach acceptance on first submission or after feedback, four end in oracle failure, none in provider unavailability — 8/12 (Wilson CI [39.1, 86.2]): RC4 0/3, ChaCha20 2/3, CBC 3/3, ECDSA 3/3; four accepted on the first submission, four after feedback, 31 submitted programs (per-task counts: Appendix E). The trajectories explain the split: the RC4 brief omits the all-zero second message that makes the XOR relation usable, ChaCha20 requires a reference-encryption observation the brief lacks, CBC shows a successful revision path, and ECDSA accepts the first submission in all three sessions.

8.2. Reduced-Information Variant

A blind-opportunity variant (containers 0032–0034) removes statement-level references: ECDSA completes 2/2 and RC4 2/2 on first submission; ChaCha20 completes 0/2 with 3 candidates over 4 responses. The reduced ChaCha20 prompt removes the statement that the reference encrypts zeros; a local solver using that zero-reference recovers both secrets, so recoverability holds when the premise is available. RC4 retains known-plaintext fields and ECDSA curve/signature observations: the information loss is asymmetric. This is an information-contract diagnostic, not a family ranking.

Answer to RQ5. The fixed executor establishes feasibility (12/12), the prospective cohort completes 8/12 with the weakness class disclosed, and the explicit-observation cohorts 6/6 per family under different contracts. The sequence locates three requirements: a usable relation, a program implementing it, and evidence binding its candidate to the verdict. Their contrasts estimate no single-requirement effect. On RQ5's two printed halves: the supplied AES-CTR chain releases its payload and four cells transfer within-family; consequence is established at the evidence level, independent discovery untested (Section 10).

9. Attribution, Scope, and Defensible Credit

9.1. Attribution Policy and Decisive Tests

Without a strict attribution policy, any claim that an autonomous agent performed an attack is unverifiable and accountability unenforceable. RQ4 specifies when an evaluator may credit a checked candidate to an execution: a signature authenticates a record, while credit additionally requires an authorized role, a reference check of the captured output, and bindings to the same attempt. The policy makes those obligations executable: an observer-signed record covers the exact executed bytes and captured stdout; a reference-signed record recomputes the task predicate from private target state against that stdout and names the observer record. Attribution requires agreeing digests, matching identity fields, authorized roles, and the replay policy; the claimant's `claims` block never influences the decision.

Conditional attribution argument. Let B contain a certificate, an observer record E_o, and a reference record E_r. The intended condition is

$$A(B) = \text{ATTRIBUTED} \implies \text{TrustedExecution}(B) \ \wedge \ \Phi(\text{Output}(B)).$$

Under Section 3.1's assumptions, the observer binds executed bytes to captured stdout; the reference recomputes Φ on that stdout from private target state. Both signed contexts must agree with the certificate's target, attempt, and epoch, carry authorized roles and kinds, and agree on source and candidate digests; the reference must also name E_o's digest, and replay handling must satisfy the declared policy. Authenticity alone leaves cross-record consistency unchecked; consistency alone leaves the issuer's authority unchecked; the argument depends on truthful trusted roles and correct reference adjudication.

Ablation evidence. Table 7 measures what each policy obligation buys: on the held-out IAR-B set, joins alone incorrectly promote 16/44 attack-family cases and signatures alone 28/44; the combination 0/44. IAR-C's preserved set gives the same qualitative separation; shared modules and calibration use limit its independence. All policy variants are author-constructed: join-only and signature-only are deliberately weak baselines designed to fail, and their failure on cases built to require both halves is a construction check, not an independent baseline — constructed-case outcomes, not deployment rates. A stronger baseline must come from outside this design. Denominator details (Table 7).

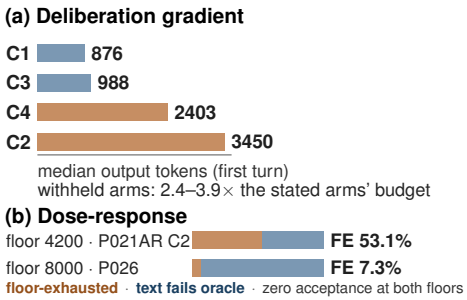


Figure 5. The deliberation-budget mechanism. (a) Median first-turn output tokens per cell in the 96-instance power cohort P021AR over text-bearing first turns (C2 n=45 of 96): the withheld arms spend 2.4–3.9× the stated arms' budget before any submission. (b) The registered floor-sensitivity cohort P026 reruns C2 on the same 96 seeds at the raised floor (4200→8000 tokens): floor-exhaustion falls from 53.1% to 7.3%, the freed sessions produce text that fails the oracle, and acceptance stays zero at both floors (generated from the locked evidence artifact).

Whether a non-zero exit may accompany a predicate-satisfying output is a declared policy choice (frozen-output vs. completion reading).

Answer to RQ4. Candidate correctness does not bind a result to its claimed execution: the ablations show why the policy needs both authorized evidence and cross-record consistency, and the mutation and ingress results examine its behavior on specified defects — support for the conditional attribution contract; they neither resolve E5's acquisition path nor establish soundness against compromised trusted roles. Section 10 gathers the five answers into one ledger.

10. Discussion

A reported attack success merges decisions by five owners: the agent's program, the evaluator's admission rule, the prompt's information contract, the session wrapper, and the compute allowance a protocol grants each turn. Behind them stands the further question: whether the evidence credits the claimed execution.

Where the nulls live. Every null among the five ownership shares lives in the ledger's Bound column (Table 4); the representation-equivalence, floor-sensitivity, and oracle-separation nulls print beside their contrasts. No null is pooled, re-tested, or averaged into a headline.

The ownership ledger. One row per owner, no pooling.

RQ1: admission, audited — the first owner we audit is our own harness. The declared evaluator retains the -0.8 point contrast; -32.8 pp is counterfactual under the earlier three-module filter, not an observed degradation: fixed programs isolate the admission decision, the unused-import diagnostic identifies its trigger, cleanup removes the negative gap, and tracing shows the trigger executes. The magnitude has no external instantiation (Section 7). The admission share is measured twice — within-harness counterfactual and external stranding — and both measurements implicate the same evaluator: ours. The frontier repanel measures our own rules on external tasks: every defined contrast is zero — the import axis never binds on frontier programs; whether a third-party rule moves its published number under regrade remains registered-open.

RQ2: the information contract owns a measured share. The premise slot owns the cell: the registered paired C1-versus-C4 contrast is one-way in all sixteen family members (fifteen Holm-significant; the 5/0 borderline prints as its own bound), naming is inert, and the share is bounded to premises about secret content (Table 3; full cohort records: Appendix E). Figure 5 shows the withholding signature before the oracle and P026's floor-sensitivity dose-response on the same 96 seeds; the caption carries both panels' numbers. The share tracks premise observability, not cipher; vendor enters where Moonshot's cohort-relative degree shows observability is not sufficient (Section 5). A pre-registered confirmatory cohort on `gpt-5.6-sol` (Holm family declared up front; the executed C1 commitment, Section 11) then bounded the share's capability dependence: C4 accepts 11/16 and the paired contrast does not clear the registered Holm bar (5/0 one-directional, p=0.0625, Holm 0.125) — the convention-known frontier model recovers without the premise, the observability axis's own prediction at frontier capability. The share falls as models become able to infer the withheld content — the estimand working: the instrument measures the premise's marginal value where the model cannot already obtain it; the method, not any magnitude, is the transferable claim.

RQ3: a bounded wrapper null, resolved at the powered size. At ten pairs, acceptance is descriptively indistinguishable (two-sided 0.65) with the direction reversed relative to the frozen cohort; post-hoc re-adjudication traces that reversal to wrapper format compliance rather than to information value (Section 6), yet the all-measures falsifier is FALSE through the revisions clause. The wrapper's only realized channel is the revision path (2 of 7), far below the registered design's power; the Fisher denominator defect is corrected and disclosed. The pre-registered powered cohort (32 pairs, label `gpt-6.1-sol`) then resolves the question at its registered bar: all 64 sessions accept on the first turn, the paired discordance is 0/32 in each direction, and the interactive arm's advantage is bounded at 10.7% (Wilson 95%) — the wrapper owns no share of this number at frontier capability.

RQ4/RQ5: consequence and attribution. The fixed executor's 12/12 operations establish feasibility; V4 achieves 8/12 agent construction; one AES-CTR chain releases a payload; four cells transfer within-family; and E5 leaves acquisition unresolved — the evidence requirements for execution credit, without reconstructing the historical path. Table 7 isolates unsupported promotions and Section 9.1 presents the conditional argument — complementary obligations that existing tools implement (the in-toto inspection). The constructed cases establish neither deployed prevalence nor independent validation; the record-serialization losses are why failure-stage retention is needed to read terminal outcomes (full records: Appendix E).

10.1. Contribution to Security-Agent Measurement

The instrument's unit is the submitted program and its execution evidence, not the benchmark task. The validity-audit literature inspects benchmark artifacts and reports defects; none of it re-scores fixed agent outputs under a declared, pre-registered, hash-frozen intervention. YHZ does: fixed-bytes regrade with output-preserving cleanup, runtime tracing, a pre-registered 2×2 slot rotation, and a falsifying family boundary run (per-item detail: Appendix E). That is the shift: a reported capability number becomes a property of the contract that produced it, auditable owner by owner; Section 13 distills the audit into five preconditions.

TABLE 4. OWNERSHIP LEDGER: ONE FROZEN INTERVENTION PER OWNER; MEASURED SHARE; BOUND.

Owner	Intervention	Registered result	Bound
Admission	regrade fixed bytes; cleanup	$-0.8/-32.8/+3.9$ pp	trace 100/101; 128/128 outputs; external 0/2 defined tasks move (two undefined; sign test p=1.0; descriptive rows in §7)
Information	2×2 prompt slots	premise stated-arm deterministic in every Claude- and GPT-family evaluable cohort (the scale-up's damaged first pass, C1 27/32 and C3 30/32, is the printed exception; Moonshot's stated arm is not, by measurement); paired one-way in all 16 family members (96/0 at n=96, Holm 4.04×10^{-28})	RC4 + ECDSA nulls; naming inert; the gpt-5.6-sol arms' C4 cells floor-exhaust at every floor tried (budget-bound coding in supplement record S30); the kimi label brackets cohort-relatively (null and collapse both measured clean)
Premise (out-of-domain)	withhold the intruder's host and cron-interval premise (non-crypto audit forensics)	8/8 stated vs 0/8 withheld oracle-correct	exact one-sided Fisher 7.8×10^{-5} at the registered n=8; broader domain coverage registered-open
Compute allowance	published protocol rerun at declared vs raised allowance	$20/50$ solved at 12,000 tokens; $28/50$ raised	$+16$ pp swing owned by the allowance; 17 slots transport-unmeasured (Appendix D)
Wrapper	interface swap	5/10 vs 7/10 (two-sided 0.65); post-hoc 8/10 vs 7/10	powered 32-pair null (0/32 each direction, advantage $\leq$10.7% Wilson 95%); falsifier FALSE via revisions (2 of 7); the measured share is diagnosed post hoc as a format-compliance effect (Section 6)
External	declared vs stress rule	31/37 vs 20/37; 12/19 correct retained; Amend3 261/282 admitted, 0 correct	F1 FALSE (0/4; defined 2; Amend3 0/3 labels); censuses 4/14, 0/40, 7/16
Attribution	join/sig/both ablation	0/44 vs 16/44, 28/44	honest 14/14; positive control

10.2. Where the Contract Concept Travels

The concept is not crypto-specific: in web exploitation the premise might be a vulnerable Apache version; in reverse engineering, that a structure sits at a fixed offset — the load-bearing sentence plus the domain's own success check. Any domain with that pair can run the same frozen 2×2 — cryptography was the first instantiation because its oracle is deterministic and byte-exact; the discipline transfers, the numbers do not, and a new domain earns a share only by running its own. The rule's first out-of-domain witness is now measured (the ledger's final row): a non-crypto audit-forensics task whose premise — the intruder's host and cron interval — selects one marker among 48 artifacts-indistinguishable decoys, and withholding it collapses acceptance 8/8 to 0/8 (exact one-sided p=7.8×10^{-5}; a first witness, with broader domain coverage registered-open).

10.3. Implications for Agent Evaluation

The YHZ floor compresses to operator practice: publish the contract the executor enforces (prompt–executor consistency checked; regrade distinguished from generation); keep source, runtime, predicate, provider, and acceptance states distinct; bind acceptance to execution; record honest nulls — Table 9 is the reviewer-side checklist.

11. Limitations, Ethics, and Reproducibility

Each limitation below names the claim it bounds and what survives it.

Scope and identity. The design prioritizes pre-registered, hash-frozen interventions over large-scale sweeps; feasibility samples are deliberately small (n=2–4; V4 n=3 per task). At n=16 per cell the cohorts reach 0.82 power for a 46-point difference (full grid: the supplementary bundle). V4 uses a mutable provider label; the post-freeze study runs another label with one experimenter; the identity-image and attempt-bound caveats are itemized in Appendix E.

New-cohort limits. The RC4 factorial is saturated (16/16 first-turn), so its honest null is not evidence of absence; no cohorts are pooled (the powered cohorts, n=8 then n=16, register the response).

The wrapper extension runs the registered minimum $n \geq 10$ pairs, where ten pairs reveal only large effects; the direction reversal is diagnosed post hoc as wrapper format compliance, not an information effect. That diagnosis rests on a post-hoc extraction rule, so the reversal stands as an instrument effect, not a resolved comparison (7 turn-1 rejections here versus 0/4 there; the unmatched-model oracle-separation control supports no cross-model comparison: Appendix E). The pre-registered powered cohort (32 pairs) then resolves the wrapper question at its bar (Section 6).

Scale and cohort size. The new-cohort limit is registered, not implicit: the smallest label base is n=16 per cell (P014, P016, P017), with n=24–96 elsewhere and a completed per-cell n=32 scale-up (roster and census: Appendix E). Seven of the twenty-four pre-specified information-contract cohorts run `claude-sonnet-4-6`; three further labels rerun the design with disclosed transport deviations (four on `gpt-6.1-sol` and `gpt-5.6-sol`, two on `gpt-6-astra`; Appendix E); two run `kimi-k3`; and the modern-AEAD wave (P014–P024) repeats the factorial on AES-GCM, ChaCha20-Poly1305, and AES-GCM-SIV, two at 96 CSPRNG-sampled instances (Section 5). Every lost session is itemized by a receipted census (246 transport-caused, 314 floor-exhausted under a 128/128 raw-body signature); every affected cohort reran clean, and three recovery passes are disclosed as author-directed substitute-label runs (`gpt-5.6-sol` after the registered label's quota exhaustion; per-group splits in Appendix D).

The cross-vendor mis-registration and withdrawal are recorded in full in Appendix E (all 97 raw responses echo the primary label).

Design resolution. Each size is stated with its exact resolution (unpaired two-sided Fisher at α=0.05, baseline p=0.5). The power at the registered 46-point difference is 0.05 at n=4 per arm and 0.47 at n=10 — sizes that cannot reliably detect even that difference; the wrapper extension and the frozen pairs can bound neither direction, while the powered 32-pair cohort bounds the interactive arm's advantage under its registered six-discordance bar. The paired

tests carrying the headline contrasts have a lower floor: exact McNemar reaches α at six one-directional discordant pairs (p=0.03125; five give 0.0625), so the P010P cohort's $5/0$ is non-significant while $10/0$, $8/0$, and $8/0$ clear it (bounded in advance, not defended: Appendix E).

Causal and measurement limits. H1-R3's conclusion covers the tested representations, levels, margin, two families, and one route: the powered $L3$ cells succeed in 124–126 of 128 (little headroom), few discordant pairs limit precision, and the bootstrap resamples this cohort only; lower-level $R2$ contrasts stay inconclusive under the frozen margin (secondary-arm detail: Appendix E). V4 and explicit-observation cohorts differ in allowance, format, parser, feedback, and observations (descriptive contrast): fixed-executor 12/12 versus V4 8/12 is not a causal gap estimate, and the supplied AES-CTR chain does not establish independent discovery. TTE_M is quantized at 0.1 s; endpoint and retention gaps: Appendix E. H1–H3 assess release-evidence linkage and audit reliability ($\kappa = 0.78$), not system correctness.

Reproducibility and artifact mapping. The reviewer bundle supplies replay code and inputs, attribution ablations, census/contrast records, and the frozen preregistrations; the external-rule regrade ships as a one-command lab re-deriving every registered and external-provenance oracle count on the frozen records, and every registered statistic regenerates from the raw per-seed rows by closed-form enumeration (Python standard library; no statistical package), checked by the evidence gate; the prior local replay agreed with all 852 verdicts; that validates the component, not the package. Full reruns require trust bundles and runtime. Floor-exhaustion bodies are logged per-session (`failure_log.jsonl`), the SOURCE_MAP pointer resolves each printed cell to its raw `results.jsonl` row, and the census population is every registered session in the frozen artifacts (Table 5); the public bundle is built by `tools/build_public_bundle.py` with its availability disclosed per release (pointers: Appendix E).

Cost. The run ledger's cost fields are estimates from the gateways' advertised rate schedules and understate the operator's actual invoicing, which totalled approximately $600 across the campaign — one gateway's invoice alone reached about $140. The per-cohort token counts in the released records are the measurement-relevant quantity.

Registered follow-up commitments. (C1) One confirmatory cohort with the Holm family declared and hashed *before* its first model call — executed (2026-11-07): the family (m=3: H1 paired primary, H2 premise marginal, H3 naming) frozen in `FREEZE_C1.json` before the first call. The registered bar was NOT cleared (H1 paired 5/0 one-directional, p=0.0625, Holm 0.125): on `gpt-5.6-sol` the registered bar did not establish the ChaCha20 zero-reference premise as load-bearing — the observability axis's own prediction at frontier capability. (C2) Two boundary-evasion operators (dynamic `exec`; submodule import) joined the mutation roster — both rejected under both arms on both bases (C2_EXECUTION.json). (C3) The nine-field release audit applied by an independent reviewer to one further team's artifact — pending the artifact choice. None changes any number here.

Unmeasured owner: memorization. The guessability axis is binned by fixture class, not measured; a model's pretraining prior over the hidden content is not separately controlled.

Ethics considerations. Targets are local, purpose-built fixtures; no production system, personal data, or human subject is involved. Escrow secrets are revealed only after a verification attempt; the safety probe reports refusals; the withdrawn composite stays withdrawn. Release risk: the public bundle holds executable recovery programs and traces for local, self-contained fixtures with no reachable production target.

Generative AI usage. Language models are the object of study: every cohort declares its model route and query budget, harness-capped (Section 6). Quantitative claims rest on retained records, frozen analyses, and task-oracle verdicts; automated checks cover arithmetic and identity links.

Scope note. The contract concept travels beyond cryptography (Section 10.2); the limits above bound the measured instances, not the concept: every measured ownership share to date lives inside a crypto oracle except the transfer witness's own; broader domain coverage remains registered-open. The regrade lab ships with the released records: a team regrading its own frozen outputs under a declared rule converts the rule-provenance instantiation into an external run (one instantiation with a collaborating research group underway).

12. Conclusion

A reported attack success has owners, and this paper measured each share. The admission rule holds a -32.8-point counterfactual whose trigger executes in 100/101 traces; the reference premise is load-bearing one-way across the registered cohorts and labels; and the observability axis reads the frontier confirmatory cohort's un-cleared Holm bar as its own prediction at frontier capability.

The wrapper question is settled at its registered size. The powered 32-pair cohort resolves the wrapper's interactive-vs-one-shot question at its registered bar: 64/64 sessions accepted turn-1, 0/32 discordances in each direction, the advantage bounded above by 10.7% (Wilson 95%). The external record leaves portability unestablished at the registered bar, as a decomposed floor: two censuses close at quorum, the ownership boundaries are named and measured (transport, normalization, admission interface, capability floor), and the frontier repanel on the visible-completion transport solves 35 of 36 sessions while its admission interface qualifies 7 of 61 and the same frozen programs score 61 of 61 under the three external teams' published graders. The normalization is one of the owners — the transport's content filter even blocks one benchmark task outright — so the verdict rests on measured contracts rather than a flat floor, and attribution is evaluated by ablations. Only the premise, naming, and derive-frame readings span five provider-attested labels across three model families, and the contract share is further bounded to premises about secret content, its degree tracking how obtainable the hidden content is from non-prompt sources (the ECDSA and RC4 nulls bracket the AES-CTR collapse).

Table 4 is the takeaway; the V1–V4 playbook (Appendix F) is the operator's side. The general claim is the method, not an external benchmark's decomposition; the ciphers are the testbed, not the thesis (Section 10.2). Commitments C1–C3 (Section 11) cover each open point (*the anonymized reviewer bundle*), and Section 13 distills the design into five preconditions — the YHZ floor: strip, declare, separate, publish, bind — as the transferable core future measurements can cite and comply with.

13. Five Preconditions for Future AI Security Measurement

The failure modes this paper measured are structural; evaluation that does not bind them in advance reproduces them. Five preconditions constitute *the YHZ floor*.

The five conditions — strip, declare, separate, publish, bind — with the demand each makes and what a compliant release shows are the reviewer-side checklist of Table 9; this paper's conformance is the worked example (Appendix E).

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Appendix A: Registered Session Inventory

The cohorts are small because each is a paired estimator sized by its registered resolution: no cohort is ever pooled. Table 5 inventories it — **3,562 agent sessions are recorded**. Each cohort count is read programmatically from the frozen artifact — a `results.jsonl` row count, except the H1-R3 campaign row, which reads its retained `DERIVED.json` session total; none is restored from a summary; the registered scale-up completed and the modern-AEAD wave (P014–P024) joined it (Section 5).

TABLE 5. REGISTERED EVIDENCE INVENTORY. SESSION COUNTS ARE ROW COUNTS OF THE FROZEN `RESULTS.JSONL` FILES PER COHORT — THE H1-R3 CAMPAIGN ROW READS ITS RETAINED `DERIVED.JSON` SESSION TOTAL, THE SAME UNIT — WITH EVERY FLAGGED INSTRUMENT-DUPLICATE ROW INCLUDED RATHER THAN SILENTLY DROPPED: SEVEN IN THE INFORMATION-CONTRACT ROW (2,112 DISTINCT SEED–CELL SLOTS) AND ONE IN THE P011-L3-N32 COUNT (128 SLOTS). EXPLORATORY PRE-PROTOCOL FEASIBILITY RUNS ARE NOT COHORT-REGISTERED AND ARE EXCLUDED.

Study family	Cohorts	Sessions	Unit
Admission counterfactual (L3)	1	852	frozen L3 pair
Information contract (premise/slot)	24	2,119	paired seed-cell
Info-contract scale-up P011-L3-N32	1	129	paired seed-cell
Wrapper / agency (H2)	3	92	seed pair
External boundary	4	370	frozen pair
Total	33	3,562	—

Notes: All rows reference retained hashes; no cohort result row was regenerated, re-labelled, or imputed to build this table. The external-boundary row's four cohorts are the initial one-session admission probe, the Amend-1 grid, and the two NYU panels. The published-protocol audit (Cohort D, Section 7) is a separate population: its 60 registered sessions are not part of these 3,562 and are never pooled with them.

Appendix B: Admission-Policy Ablation

Table 7 ablates the combined attribution policy on the constructed attack-family cases: join-only evidence does not accept full attributions (10/14, 10/14, 14/16 honest claims

TABLE 6. H2 WRAPPER OUTCOMES. FROZEN (SEEDS 61–64), EXTENSION (SEEDS 65–74), AND POWERED (SEEDS 441–472) ARE SEPARATE REGISTERED COHORTS; NEVER POOLED.

Cohort	A acc.	B acc.	A rej.	A rev.	A rec.
Frozen (61–64)	4/4	2/4	0	0	0
Extension (65–74)	5/10	7/10	7	3	2
Powered (441–472)	32/32	32/32	0	0	0

Notes: Columns: accepted sessions per arm, A turn-1 rejections, A turn-2 submissions, A turn-2 recoveries. B has no turn 2 structurally.

TABLE 7. RQ4 ABLATION: UNSUPPORTED FULL-ATTRIBUTION PROMOTIONS ON CONSTRUCTED ATTACK-FAMILY CASES. FRACTIONS USE EACH SET'S OWN ATTACK-FAMILY DENOMINATOR, INCLUDING MALFORMED CONTROLS. HONEST CLAIMS ACCEPTED BY THE COMBINED POLICY ARE COUNTED SEPARATELY AT THEIR SUPPORTED CLAIM LEVEL.

	Unsupported promotions			Honest
Set	Joins	Signatures	Both	Both
IAR-B dev	16/40	24/40	0/40	14/14
IAR-B held-out	16/44	28/44	0/44	14/14
IAR-C preserved	10/28	14/28	0/28	16/16

Notes: Join-only does not accept 10/14, 10/14 and 14/16 honest claims, respectively: its full-attribution-only contract excludes lower-level claims. Those counts are not evidence that an otherwise supported full-attribution claim was rejected. Sets are not pooled.

TABLE 8. ORACLE-LEVEL REPRESENTATION CONTRAST ON THE EXTERNAL GRID (DESCRIPTIVE; NO ADMISSION RULE APPLIED). TURN 2 RESCUED 0 SESSIONS.

Task	R0	RT	Δpp	Failure pattern
Dynastic	4/4	4/4	0	—
Makeshift	4/4	0/4	−100	transport-encoding conflation
Primary Knowledge	3/4	4/4	+25	trial-division timeout
Partial Tenacity	0/4	0/4	0 (0/0)	timeouts / give-ups

Notes: Descriptive only: sign test across tasks 1+/1−/2 ties, $p = 1.0$; reduced power at four tasks.

excluded by its own contract), signature-only accepts unsupported promotions (24/40 on the development set; 28/44 held-out and 14/28 preserved), and the combined policy accepts every honest claim (14/14, 14/14, 16/16) while rejecting every unsupported one — the policy needs both authorized evidence and cross-record consistency. Sets are never pooled; per-record verdict rows ship in the supplementary bundle.

Appendix C: External-Boundary Qualification

The external grid's qualification record: the frozen boundary clause rejected 52 of 57 programs on syntax alone (encoding-keyword and variable-bound opens), and the registered v5 repair (zero model calls, both verdicts printed) qualifies 55, after which every defined contrast is zero — the import axis never binds on frontier programs (Section 7). Table 8 carries the oracle-level representation contrast on the four qualifying tasks; per-program verdict rows and both repair verdicts ship in the supplementary bundle.

Appendix D: Cohort Records

P010 discovery-cohort record (Section 5). On ChaCha20 (P010, 16/16 sessions, 0 provider failures, 22 responses), the cell collapses under exactly one slot combination — name stated, premise withheld. The premise marginal is 8/8 with the reference stated against 4/8 without it (Fisher one-sided $p = 0.03846$, two-sided $p = 0.07692$); the naming marginal shows $p = 1.0$, its apparent contrast sitting entirely in C4; the paired C1-versus-C4 discordance is 4/0 on the shared seeds (binomial two-sided $p = 0.125$). Every acceptance under a stated premise succeeds on the first turn (8/8 across both naming conditions). The frozen falsifiers read: F1 FALSE, naming plus an unstated reference fails the cell outright (C4 0/4); F2 FALSE, withholding the name does not gate success while the premise is stated (C3 4/4, first-turn); F3 TRUE, the derive-from-observations frame succeeds without the sentence (C2 4/4, at higher submission cost). Retained failure stages: C4 loses 7 sessions to oracle rejections plus 1 to an execution/parse failure; C2 loses 1 and 1 (per-seed rows in Appendix D).

RC4 honest-null record (Section 5). The structural replication on RC4 (P011) returns the anticipated honest null: all 16 sessions succeed on the first turn; C4, the cell that collapsed on ChaCha20, completes 4/4; both marginals sit at $p = 1.0$; the paired discordance is 0/0 and not evaluable (F1 TRUE, the honest empty premise). The code route recovers via $c_{\text{target}} \oplus c_{\text{ref}}$ under an assumed zero reference (14 of 16 stating that assumption explicitly); a retained underdetermination witness shows the instance is not solvable from the public fields alone — three distinct reference values each yield a distinct consistent candidate — so the success came from the model's standing assumption, not information the prompt supplied. P011 record (Appendix D).

The per-cohort records below are quoted in full from the main text's earlier revisions; every number is frozen in its cohort artifact.

A powered same-label replication (seeds 101–108, per-cell n=8, frozen runner byte-identical) tests the pattern at power; it ran after external review set the per-cell n=8 power requirement against cohorts whose outcomes were already known. C1 and C3 saturate (8/8 each, first-turn); C4 drops to 3/8 and the derive frame C2 collapses (0/8, all reaching turn 2 and failing). The pre-specified and hash-frozen primary test is the paired C1-versus-C4 discordance on shared seeds. It reads 5/0, all C1 (two-sided $p = 0.0625$). The premise marginal reaches conventional significance (16/16 vs 3/16, Fisher exact one-sided $p = 1.6 \times 10^{-6}$, two-sided $p = 3.2 \times 10^{-6}$; an earlier internal estimate of p=0.00055 did not reproduce under the exact test and is corrected here, frozen counts unchanged). F1 FALSE (the honest null does not replicate; the premise effect does), F2 FALSE, F3 FALSE (the derive frame's 4/4 does not survive power).

P010P record (Appendix D).

A per-cell n=16 cohort (seeds 121–136, same label, frozen runner byte-identical, P010N16) was pre-registered after the external review's statistical requirement: C1 and C3 saturate again (16/16 each, first-turn); C4 reads 6/16 (five first-turn acceptances, one at turn 2) and the derive frame 3/16 (29 submissions; 13 sessions consumed turn 2 and failed). The pre-specified and hash-frozen paired C1-versus-C4 test reads 10/0, all C1 (six ties; binomial two-sided $p = 0.001953$); the premise marginal is 32/32 versus 9/32 at n=16 (Fisher one-sided $p = 1.9 \times 10^{-10}$, two-sided $p = 3.8 \times 10^{-10}$; seed-stratified test ($\chi^2 = 28.85$, $p = 7.8 \times 10^{-8}$, post hoc); the naming marginal stays inert (22/32 versus 19/32, two-sided $p = 0.6029$). F1 FALSE (the honest null is refuted one-way at n=16), F2 FALSE (C3 equals C1), F3 FALSE (3/16 far below 16/16). The withheld arm's degree moves across same-label cohorts (C4: 0/4, 3/8, 6/16) while the stated arm stays deterministic (56/56 across the three same-label cohorts). P010N16 record (Appendix D).

A second-label replication (per-cell n=8, seeds 111–118, `gpt-6.1-sol`, four disclosed transport deviations,

P010L2) tests which shares are label relative. The premise share transports and strengthens (16/16 with it stated against 0/16 withheld (Fisher one-sided $p = 1.7 \times 10^{-9}$, seed-stratified test ($\chi^2 = 24.0$, $p = 9.6 \times 10^{-7}$, post hoc); paired C1-versus-C4 discordance 8/0 (two-sided $p = 0.0078$))). Naming stays inert (8/16 vs 8/16 ($p = 1.0$)) and the derive frame stays refuted (C2 0/8); only C4's degree differs by label (0/4, 3/8, 0/8) — the second label changes the serving stack, not the model claim (Section 11). P010L2 record (Appendix D).

A third-label replication (per-cell n=8, seeds 201–208, `gpt-6-astra`, same route, same four transport deviations, P010L3) tests whether label 2's split was a property of that pair of routes. It is not: C1 and C3 saturate again (8/8 each, first-turn), and C4 and the derive frame C2 are 0/8 again (16 submissions each, every session consuming turn 2 and failing). The premise marginal repeats the same split exactly — 16/16 against 0/16 (Fisher one-sided $p = 1.7 \times 10^{-9}$, two-sided $p = 3.3 \times 10^{-9}$; seed-stratified $\chi^2 = 24.0$, $p = 9.6 \times 10^{-7}$, post hoc), with paired C1-versus-C4 discordance 8/0 (two-sided $p = 0.0078$) and naming inert at 8/16 versus 8/16 ($p = 1.0$). P010L3 record (Appendix D).

A fourth-label replication (per-cell n=8, seeds 211–218, `gpt-5.6-sol`, P010L4) first ran on a route that degraded mid-run (14 of 32 sessions lost, 14/14 withheld); its registered clean rerun measures the full design: the stated arm is first-turn deterministic (C1 and C3 8/8; 16/16), while the withheld arm reads 2/16 accepted (C2 2/8 — the two acceptances the damaged pass glimpsed are real on this label — with the other six C2 sessions and all eight C4 sessions floor-exhausted under the 2800-token floor), paired C1-versus-C2 6/0 (two-sided $p = 0.031$) and C1-versus-C4 8/0 ($p = 0.0078$) under the non-acceptance coding; under the registered evaluable-pairs rule the statistic holds no evaluable C4 pair — all eight C4 sessions are floor-exhausted — and the registered statistic is not evaluable for this cohort. C4's rejection degree is measured on the repaired transport, not observed to collapse. P010L4 record (Appendix D).

A fifth-label cohort (P010L5+L5R, seeds 221–228 and 231–238, `kimi-k3` on Moonshot AI via a commercial gateway route, a different model family and provider) reveals a qualitative boundary, now measured twice on clean transport: the stated arm is not deterministic on this label (the only label where this happens), and the two seed cohorts bracket the label's behavior — L5's clean rerun replicates the null (stated 9/16 vs withheld 6/16, Fisher two-sided $p = 0.48$: the first pass's null was real, not a route artifact), while L5R's clean rerun replicates the collapse (stated 7/16 vs withheld 1/16, $p = 0.037$; first pass 11/16 vs 2/16, $p = 0.0032$). The contract share replicates on Claude and GPT families; on Moonshot the effect is cohort-relative, as the degree claim states. P010L5+L5R record (Appendix D).

A fresh-seed n=16 cohort (P010X1, seeds 161–176, primary label) imports the frozen runner byte-for-byte: the paired C1-versus-C4 contrast reads 10/0 with six ties (two-sided $p = 0.001953$), the premise marginal 32/32 versus 12/32 (Fisher two-sided $p = 2.3 \times 10^{-8}$; CMH $\chi^2 = 25.00$, post hoc), naming stays inert ($p = 1.0$), and the stated arm is again first-turn deterministic. The withheld arms land at 6/16 each: fresh seeds move the mutable arms' level, not the share (Figure 4); the one-way discordance repeats. A provenance correction attaches: registered as cross-vendor (`gpt-oss:120b-cloud` via the local gateway), its label redirect never bound at runtime and all 97 raw responses echo the primary label; the vendor claims are withdrawn and the cross-vendor question returns to open (Section 11). P010X1 record (Appendix D).

D.3 External-boundary records. The census: Cybench 14 crypto tasks (4 qualifying below quorum; 0 of 40 CVE-Bench tasks), then NYU CTF Bench (7 of 16 candidates meet quorum). The Amendment-3 panel: 168 sessions × 3 labels × 282 programs, 261/282 admitted after the repaired read boundary, none oracle-correct under the registered whole-stdout rule — F1 FALSE on all three labels; the post-hoc decomposition attributes the null to named owners (8 sessions whose submitted output printed the flag, rejected by the answer normalization — 11 were flag-bearing at some turn, three losing it to a crashed revision; 103 terminal crashes hidden by a status taxonomy with no crash class; 45 terminal outcomes without a crash or a flag); the RT delivery record is byte-exact 84/84 (the earlier "broken delivery" was an audit-regex defect, withdrawn). *The Amendment-5 transport-repaired repanel — the record behind the abstract's anchor.* Three frontier labels (`gpt-5.6-sol`, `gpt-6-astra`, `gpt-6.1-sol`) on the OpenAI-responses transport, 9 census-pass (task, representation, label) combinations × seeds 425–428 = 36 sessions, hash-frozen before the first call; 35 of 36 sessions oracle-correct under the registered whole-stdout rule (eps 23/24; lowe R0 12/12), while the frozen admission interface qualifies 7 of 61 retained programs (the argv1 boundary clause rejecting 34 of the 35 oracle-correct programs); the registered falsifier family prints F1 FALSE, F2 TRUE, F3 PASSED_preservation (61/61). The Amendment-4 facade record and the repanel's full narrative are in Section 7. Per-session rows, per-label grids, and the audit's finding records ship in the supplementary bundle.

Appendix E: Wave and Disclosure Records

Import-filter regrade detail (Section 4). Level composition: XOR uses 2/4/8 ciphertexts with 0/1/3 decoy cribs; ECDSA uses 2/6/16 signatures with one repeated-nonce pair; L3 additionally requires the nonce and scalar and a 128-byte XOR target. The 205 denied-module sessions split into 183 of the 768 representation trials and 22 of the 84 control sessions. The paired bootstrap over the fixed 128-instance L3 set (50,000 draws, seed 20261021; instances resampled within family, families weighted equally) gives an unadjusted 95% interval $[-42.2, -21.9]$ points for the -32.0-point regrade change. Trace coverage: 24 matched L3 instances, 77 elsewhere; the remainder pre-date per-session tracing.

Modern-AEAD wave narration and family correction (Section 5). *Family-level correction over the chain's primary statistics.* Because each cohort was registered after its predecessor's outcomes, inference is anchored to the registered family, not to any single cohort. The family comprises the 16 evaluable paired C1-versus-C4 registrations completed after the 2026-11-03 transport repair whose registered primary statistic is informative: every member holds at least five evaluable pairs. Its membership is explicit. The ChaCha20 chain contributes five completed cohorts: the powered P010P, the n=16 P010N16, the withdrawn-vendor P010X1, the second-label P010L2 rerun, and the third-label P010L3. The AES-CTR boundary run P013 and the scale-up rerun P011-L3-N32 follow. The modern-AEAD wave contributes nine: P014, the P016 `gpt-6.1-sol` arm, P017, P018 (its registered-label first pass; the recovery rerun is label-split and disclosed per label above), P019, the

power cohort P021AR, the P022a cross-label arm, P023, and P024a. Excluded by design: the discovery cohort P010 and the saturated/observable controls (P011, P012, P020). Excluded under the stated rules: the P016 `gpt-5.6-sol` and P022b arms (no evaluable C4 pairs; floor-exhausted; reported by their budget-bound sensitivities), the P010L4 rerun and P024b's completed registration (each holds a single evaluable pair — see the boundary conditions below), and the kimi label's clean reruns (outside the registered family; their marginal statistics are printed in full in the fifth-label paragraph above, paired records in Appendix D). *The family at one glance.* Table 3 carries the sixteen members' registered statistics: the paired C1-versus-C4 discordance, its exact two-sided binomial p, the step-down Holm adjustment at FWER 0.05, and the Clopper–Pearson two-sided 95% lower bound on the one-directional discordance probability. Every cohort that stood alone at α=0.05 still clears the correction; the one that did not — the $5/0$ borderline, Holm 0.0625 — remains non-significant and prints as its own bound. Two boundary conditions sit outside the family under the same strict rule, disclosed rather than absorbed: P024b's completed registration holds a single evaluable pair (1/0, p=1.0; its first pass read 3/0, p=0.25) — a degenerate statistic produced by 46/48 floor exhaustion, not a replication measurement — while the P010L4 rerun holds no evaluable C4 pair (all eight C4 sessions floor-exhausted by transport collapse) and its registered statistic is not evaluable. Both are printed in full above and reported as consistent, under-powered at every coding. The lesson belongs to the floor itself: when the infrastructure collapses, the ability to measure fails with it — and that failure is itself evidence of how fragile evaluations are that never track such collapses. The membership rule is explicit (registration-complete replications; the discovery cohort and the saturated/observable controls excluded by design; floor-exhausted-only arms and the kimi reruns excluded under the rules stated above), and every column is regenerated from the raw per-seed records by one script at three significant figures under a single rounding rule. At the exact test's resolution floor at n=16, the smallest attainable two-sided exact p with six ties is 2^{-9} — the value the fresh-seed cohort realizes.

Discussion RQ2 per-cohort record (Section 10).

RQ2: the information contract owns a measured share. The share is the disclosure of one sentence: the paired C1-versus-C4 contrast reads 10/0 at per-cell n=16 (two-sided $p = 0.001953$), never failed where the premise is stated (5/0 in the powered cohort, $p = 0.0625$; 8/0, $p = 0.0078$, on each of two further labels). Descriptively, 56/56 with the reference stated against 16/56 without, same-label (seed-stratified CMH $p = 1.5 \times 10^{-12}$; a post hoc stratified summary, not an inference under the no-pooling rule), and 16/16 against 0/16 on each of the other labels (2/16 on the last label's clean rerun, which re-measures the 14 withheld sessions the damaged run lost: 2 accepted, 14 floor-exhausted). The derive frame collapses outside the original cohort: 0/8 at n=8 on both further labels with evaluable withheld arms and 3/16 at n=16; the fourth label's withheld arms are now measured on the repaired transport. On fresh seeds at n=16 the paired contrast repeats at 10/0 ($p = 0.001953$) with premise 32/32 vs 12/32. An ECDSA control whose premise is readable off the public observations returns the pre-registered null (32/32 vs 32/32); AES-CTR, a second hidden-premise family, returns both withheld arms at 0/16. The share tracks premise observability, not cipher; vendor enters where Moonshot's cohort-relative degree shows observability is not sufficient (Section 5). The withholding signature is measurable before the oracle sees anything (Figure 5a): in the 96-instance power cohort P021AR the median first-turn output tokens over text-bearing first turns run C1 876 $<$ C3 988 $<$ C4 2403 $<$ C2 3450 (C2 n=45 of 96; 51 of the 96 C2 sessions floor-exhaust at the 4200-token floor) — the withheld arms spend 2.4–3.9$\times$ the stated arms' first-turn budget before any submission. P026 measures the mechanism directly (Figure 5b): on the same 96 seeds at the doubled floor, floor-exhaustion falls from 53.1% (51/96 at 4200) to 7.3% (7/96 at 8000) and the freed sessions produce text that fails the oracle — withholding buys deliberation, and no floor converts it into acceptance.

Census receipts. Every lost session is itemized by a receipted census: 246 transport-caused and 314 floor-exhausted sessions under a 128/128 raw-body signature; every affected cohort reran clean, and three recovery passes are disclosed as author-directed substitute-label runs (`gpt-5.6-sol` after the registered label's quota exhaustion; per-group splits ship in the supplementary bundle).

Provenance correction (P010X1). Registered as cross-vendor (`gpt-oss:120b-cloud` via a local gateway), the label redirect never bound at runtime: the runtime metadata and all 97 raw responses echo the primary label; the vendor claims are withdrawn and the cross-vendor question returns to open (Section 11). The withdrawal record (`PROVENANCE_CORRECTION.md`) ships with the cohort artifact.

Wrapper session detail (Section 6). The five Afailures: four are turn-2 records with `stage=parse`, `submitted=false` (seeds 70, 71, 72, 74); the fifth (seed 73) emitted an envelope and crashed at execution. The post-hoc re-adjudication recovered programs from all four parse failures (seeds 70, 72, 74 accepted; 71 rejected): post-hoc acceptance 8/10 against B's 7/10, the format-compliance diagnosis. The oracle-separation control (matched arm, seeds 65–74, `oracle.query` always `False`) ran on a single substituted model: acceptance 0/10 with 10/10 evaluable and 256 total oracle calls; attrition dominates (the informative cell is n=1). Per-pair records and stored raw bodies ship in the supplementary bundle.

Attrition-concentration defect. Provider failures concentrate in withheld cells (P017's frozen attribution rule: stated-vs-withheld attrition Fisher p=0.002, budget hypothesis fails; P021 reproduces the pattern at scale, 73% of withheld-arm sessions lost versus 45% stated-arm). Flagged instrument-duplicate rows are retained, never silently dropped: one each in P014 and P021, seven in the information-contract census row (2,112 distinct seed–cell slots), one in the P011-L3-N32 count (128 slots).

Units, endpoints, and evidence tiers. Causeattribution uses record-level denominators where record and program units differ (the 27 ECDSA failure records: the scalar is miscomputed in 24 — 20 wrong-literal plus 4 undetermined). TTE_M is quantized at 0.1 s; some sessions lack raw endpoints, and legacy records breach the 1 ms gate. Evidence tiers (exploratory, V-tier verified, E-tier host-confounded) are stated per record; the cross-host timing ratio is host-confounded and exact key recovery stays unresolved. Clocks and environment manifests per cohort ship in the supplementary bundle.

Response-only information contract (H1 V2). Ninety provider responses all delivered with a recorded `end_turn`; the parser accepts 81 of 90, and 51 of the 60 sufficient-condition responses are candidate-bearing; every parser-valid

response contains the requested field (the remaining 9 fail the parser).

Consequences of evaluator-boundary defects. The H1 V2 scorer mismatch, the E5 verifier-identity mismatch, the Cybench conditional answer-flow, and the post-freeze leak gates are related verification-authority risks; no shared root cause is asserted. Three post-freeze tool fixes disclosed: the H2 Fisher denominator corrected from $\mathrm{comb}(2n, n)$ to the margin-conditioned $\mathrm{comb}(2n, T)$ (the published one-sided $6/70 = 0.0857$ corrected to $6/28 = 0.2143$, scipy-verified, frozen namespace unmodified); the P011 leak-scan reporting layer amended after freeze with both SHA-256 values retained (`FINAL_FREEZE.json`, `integrity.post_freeze_tool_amendments`); the amendment-1 regrade harness corrected offline between its freeze and outcomes commits. No experiment input changed in any fix.

The cohort registered as cross-vendor delivered the primary label at runtime (Section 5), so the labels vary the serving route and not a verified set of weights. Across the Claude- and GPT-family routes measured, premise, naming, and derive-frame readings replicate (on Moonshot's label the premise effect is cohort-relative), while C4's degree varies by label and is now measured on every label: the last label's own record holds 8/8 judged C4 sessions (0 accepted), and the rerun wave adds measured C4 outcomes on the remaining labels (all 0 accepted; per-label floor-exhausted counts in the evidence artifact). All routes are provider attestations, not weight-digest-pinned images; the raw provider bodies are released so this attestation is checkable, not trusted: the fingerprint audit that withdrew one cohort's label claim (Section 5) runs on the released files; replicating any cohort is a bounded protocol (one frozen runner, eight seeds). The P011 mechanism is model-relative, and a non-standard reference convention remains the registered next control. The claim is bounded to premises about secret content: the ECDSA control's premise is readable off the public observations and returns the pre-registered null (32/32 withheld), while the AES-CTR second family returns both withheld arms at 0/16 — the contract share is fixture-relative in degree and premise-kind-relative in kind. The censuses describe the pinned snapshots only; the initial regrade was degenerate (0/37 admissible under any rule) until the registered read-boundary repair, after which the admission mechanism runs on external programs but remains unestablished, not refuted; CVE-Bench supplied 0 crypto tasks, and replanning needs a registered amendment.

Audit worked incident. The attribution audit's worked-incident (the unsupported promotion caught by the combined policy) is in-record with its ablation row (Table 7); the per-record trace ships in the supplementary bundle.

TABLE 9. THE YHZ FLOOR AS A REVIEWER-SIDE CHECKLIST. EACH CONDITION, THE DEMAND IT MAKES, AND WHAT A COMPLIANT RELEASE SHOWS; THE ANCHORS ARE THIS PAPER'S OWN MEASURED INSTANCES.

Condition	What to demand	A compliant release shows
Strip the task	a stripped twin of every reported number, paired on shared seeds	paired C1-versus-C4 contrast per cohort (96/0) with its budget-bound C1-versus-C2 sensitivity (96/96), both $p{=}2.5\times10^{-29}$
Declare the rule	the acceptance rule frozen before the first call, measured under every admissible coding	both codings printed for P024b (1/0, $p{=}1.0$; budget-bound 24/0)
Never pool	per-group reads for populations differing in one controlled attribute	P018 22/0 registered-label read; P023 83/0 and 11/0 per label
Publish raw	per-seed rows, failure bodies, receipted censuses	one-script regeneration of every inferential cohort statistic
Bind to negatives	registered nulls stated as the claim's boundary	GCM-SIV accepts nothing in any cell; P026 dose-response 53.1%→7.3%

Appendix F: Verifying a Vendor's Security-Agent Claim

The claim is a contract; this playbook is what a reviewer can run to check it without trusting the vendor. (V1) The worked incident is in-record — a cohort registered as cross-vendor (a `gpt-oss` label through a local gateway) whose label redirect never bound at runtime, all 97 raw responses echoed the primary Claude label — so the claim was withdrawn, a substitution no summary rate can surface. Ask for the raw provider bodies first. (V2) Run the nine-field release audit on the retained outputs; it splits the reported rate into program, evaluator, and contract ownership. (V3) Ask at which size the claimed margin was detectable: a pooled rate from cohorts below their registered resolution (the D.4 record, released supplement) covers an uncertified difference. (V4) Require the admission rule and failures two-sided: a vendor median absorbs 14 no-body provider events invisibly. Credit no execution without observer-signed bytes, a private-state reference check, agreeing digests, authorized roles, and no weight on the claimant's `claims` block.